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// BootRunner.cs
using System.Collections;
using Sy.Core;
using UnityEngine;
using UnityEngine.SceneManagement;
using UnityEngine.UI;
using Logger = Sy.Core.Logger;
namespace Sy.P06.Boot
{
    public static class SceneIds
    {
        public const string Boot = "S00_Boot";
        public const string Main = "S06_Main";
    }
    /// <summary>
    /// S00_Boot      Mock  →  →      T001④
    ///      Boot      UI T001b  Boot      =
    /// </summary>
    public sealed class BootRunner : MonoBehaviour
    {
        IEnumerator Start()
        {
            BootServices.RegisterAll();
            BuildLoadingBadge();
            var sequence = new BootSequence(SwitchToMain);
            if (!sequence.RunAll())
            {
                Logger.Error("BOOT", "boot sequence failed - stay on S00_Boot");
                yield break;
            }
            yield return null;
        }
        /// <summary>S00_Boot is a bare frame while the seven steps run and
        /// the main scene streams in. The koi+crane signature mark (X1706
        /// icon) rides the bottom-right corner for that window as the
        /// loading badge: a flourish that never blocks boot, skipped when
        /// art is absent (null = no dead UI), destroyed with the scene it
        /// lives in.</summary>
        static void BuildLoadingBadge()
        {
            var art = ArtRuntime.Get("P06_icon_symbol_koi_crane");
            if (art == null) return;
            var canvasGo = new GameObject("LoadingBadgeCanvas", typeof(Canvas));
            var canvas = canvasGo.GetComponent<Canvas>();
            canvas.renderMode = RenderMode.ScreenSpaceOverlay;
            var badgeGo = new GameObject("LoadingBadge", typeof(RectTransform), typeof(Image));
            badgeGo.transform.SetParent(canvasGo.transform, false);
            var rt = (RectTransform)badgeGo.transform;
            rt.anchorMin = rt.anchorMax = new Vector2(1f, 0f);
            rt.anchoredPosition = new Vector2(-90, 90);
        }
    }
}
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        rt.sizeDelta = new Vector2(96, 96);
        var img = badgeGo.GetComponent<Image>();
        img.sprite = art;
        img.preserveAspect = true;
        img.raycastTarget = false;
    }
    bool SwitchToMain()
    {
        StartCoroutine(LoadMainRoutine());
        return true;
    }
    IEnumerator LoadMainRoutine()
    {
        var op = SceneManager.LoadSceneAsync(SceneIds.Main, LoadSceneMode.Single);
        while (op != null && !op.isDone) yield return null;
    }
}

// BootSequence.cs
using System;
using System.Collections.Generic;
using Sy.Core;
using Sy.Platform;
namespace Sy.P06.Boot
{
    /// <summary>
    /// Boot    T001
    /// ①Logger→②→③Config→④Save→⑤ILocale→⑥Ads.Preload→⑦    [BOOT] n/7
    /// ⑦    Play    =SceneManager.LoadSceneAsync Editor dry-run=EditorSceneManager.OpenScene
    /// X006    SY MOCK
    /// </summary>
    public sealed class BootSequence
    {
        public const int TotalSteps = 7;
        readonly Func<bool> _switchScene;
        readonly List<string> _lines = new();
        public BootSequence(Func<bool> switchScene)
        {
            _switchScene = switchScene;
        }
        public IReadOnlyList<string> Lines => _lines;
        public bool RunAll()
        {
            bool ok = true;
            ok &= Run(1, "Logger", LoggerStep);
            ok &= Run(2, "Time", TimeStep);
            ok &= Run(3, "Config", ConfigStep);
            ok &= Run(4, "Save", SaveStep);
            ok &= Run(5, "ILocale", LocaleStep);
            ok &= Run(6, "Ads.Preload", AdsStep);

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        ok &= Run(7, "SwitchMain", () => _switchScene != null && _switchScene());
        return ok;
    }
    bool Run(int n, string name, Func<bool> step)
    {
        bool result;
        try
        {
            result = step();
        }
        catch (Exception e)
        {
            Logger.Error("BOOT", $"step {n} {name} exception: {e.Message}");
            result = false;
        }
        string line = $"[BOOT] {n}/{TotalSteps} {name} {(result ? "ok" : "FAIL")}";
        _lines.Add(line);
        Logger.Info("BOOT", line);
        return result;
    }
    static bool LoggerStep() => true; // JS sink platform T001c
    static bool TimeStep()
    {
        var time = ServiceLocator.Get<ITimeService>();
        return time.ServerNowMs() > 0 && !time.TimeCheated();
    }
    static bool ConfigStep()
    {
        var config = ServiceLocator.Get<IConfigService>();
        var update = config.CheckUpdate("mock://manifest");
        return update.NewVersion != null; // mock= X009 manifest
    }
    static bool SaveStep()
    {
        var save = ServiceLocator.Get<ISaveService>();
        save.Save("boot_probe", "{}");
        return !string.IsNullOrEmpty(save.Load("boot_probe"));
    }
    static bool LocaleStep()
    {
        var locale = ServiceLocator.Get<ILocaleService>();
        return !string.IsNullOrEmpty(locale.GetFontSet().NormalFont); // +
    }
    static bool AdsStep()
    {
        ServiceLocator.Get<IAdsService>().Preload("boot_warmup");
        return true;
    }
}

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// BootServices.cs
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Sy.Core;
using Sy.Platform;
using UnityEngine;
using Logger = Sy.Core.Logger;
namespace Sy.P06.Boot
{
    /// <summary>
    ///     Mock     SY MOCK     X006     Mock X002a     AppID touristappid
    /// T001c WX/TT     Channel     Editor     Boot
    /// </summary>
    public static class BootServices
    {
        public static void RegisterAll()
        {
            if (ServiceLocator.TryGet<ITimeService>(out _)) return; //
            ServiceLocator.Register<ITimeService>(new MockTimeService());
            ServiceLocator.Register<IConfigService>(new MockConfigService());
            ServiceLocator.Register<ISaveService>(new MockSaveService());
            ServiceLocator.Register<ILocaleService>(new MockLocaleService());
            ServiceLocator.Register<IAdsService>(new MockAdsService());
            ServiceLocator.Register<IPlatformService>(new MockPlatformService()); // T015-A share seam
            ServiceLocator.Register<ILaunchInfo>(new MockLaunchInfo()); // T015-B return seam (null query = orga
            ServiceLocator.Register<IAlbum>(new MockAlbum()); // T015-C album seam: mock records save paths; the
            Logger.Info("BOOT", "mock services registered (SY MOCK)");
        }
    }

    /// T015-A mock platform seam (SY MOCK): editor/dev sessions expose the
    /// share face so the scent-card button is real; the G4 platform bridge
    /// replaces this registration without touching any runtime call site.
    public sealed class MockPlatformService : IPlatformService
    {
        public Channel Channel => Channel.Editor;
        public System.Threading.Tasks.Task<AuthResult> Login()
            => System.Threading.Tasks.Task.FromResult(new AuthResult { Ok = true });
        public void Share(SharePayload payload)
        {
            Logger.Info("SHARE", "mock share captured: title=" + payload.Title + " image=" + payload.ImageUrl);
        }
        public UserInfo GetUserInfo() => new UserInfo();
    }

    public sealed class MockTimeService : ITimeService
    {
        public TimeSource Source => TimeSource.Mock;
        public long ClientNowMs() => DateTimeOffset.UtcNow.ToUnixTimeMilliseconds();
        /// <summary>>SY MOCK X006 TCB serverTime→HTTPS Date → T001b </summary>
        public long ServerNowMs() => ClientNowMs();
    }
}

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        public bool TimeCheated() => Math.Abs(ServerNowMs() - ClientNowMs()) > 120_000;
    }
    public sealed class MockConfigService : IConfigService
    {
        public T GetTable<T>(string tableId) where T : class
            => throw new KeyNotFoundException("mock config table not loaded: " + tableId);
        public bool TryGetTable<T>(string tableId, out T table) where T : class
        {
            table = null;
            return false;
        }
        public ConfigUpdateResult CheckUpdate(string manifestUrl)
            => new ConfigUpdateResult(false, "mock-0.0.1");
        public void SwapPolicy(ConfigSwapPolicy policy) { }
        public string PinConfigVersion(string scope) => "mock-0.0.1";
    }
    public sealed class MockSaveService : ISaveService
    {
        const string Prefix = "sy_save_";
        public void Save(string slot, string payloadJson) => PlayerPrefs.SetString(Prefix + slot, payloadJson);
        public string Load(string slot) => PlayerPrefs.GetString(Prefix + slot, string.Empty);
        /// <summary>X006          T001b v1→v2          </summary>
        public SyncResult CloudSync() => new SyncResult(false, null);
    }
    public sealed class MockLocaleService : ILocaleService
    {
        public void SetLang(string lang) { }
        /// <summary> key key          T001b cfg_i18n + NUnit          </summary>
        public string GetText(string key, params string[] args)
        {
            Logger.Warn("LOCALE", "missing i18n key (fallback): " + key);
            return key;
        }
        public FontSet GetFontSet()
            => new FontSet("LegacyRuntime.ttf", "LegacyRuntime.ttf", "LegacyRuntime.ttf");
    }
    public sealed class MockAdsService : IAdsService
    {
        public event Action<AdLifecycle> OnAdEvent;
        public Task<RewardResult> ShowRewarded(string pointId)
            => Task.FromResult(new RewardResult { Completed = true, Reason = "mock" });
        public void ShowInterstitial(string pointId) { }
        public void ShowBanner(string pointId) { }
        public void Preload(string pointId)
        {
            Logger.Info("ADS", "preload (mock): " + pointId);
            OnAdEvent?.Invoke(new AdLifecycle(AdLifecycleType.Fill, pointId, "mock"));
        }
    }
}

```

```
// MainUI.cs
using System;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SceneManagement;
using UnityEngine.UI;
using Sy.Core;
namespace Sy.P06.Newbie
{
    public static class NewbieSceneIds
    {
        public const string Main = "S06_Main";
        public const string Battle = "S06_Battle";
    }
    /// <summary>S06          →S06_Battle M5          —
    /// W11b offline tip counter: settles the away-span into tips when returning
    /// to the main screen (parallel claim, clock never resets, cfg_idle rate law).
    /// The panel only appears when tips are pending - fresh installs and pre-boot
    /// states show nothing (no fake tips, no dead UI; the panel is pure idle
    /// economy, not an ad slot).</summary>
    public sealed class MainUI : MonoBehaviour
    {
        TipCounter _tipCounter;
        ITimeService _clock;
        Sy.Telemetry.TelemetryService _shareTelemetry; // T015-B: card_opened capture
        LetterPanel _letterPanel; // T022: guest mailbox (weekly letters)
        Text _lettersBadge; // T022: unseen-mail count on the mailbox button
        AboutPanel _aboutPanel; // T024: compliance disclosure surface
        public void StartBusiness()
        {
            SceneManager.LoadScene(NewbieSceneIds.Battle);
        }
        void Awake()
        {
            Application.targetFrameRate = 30;
            MainArtInjector.Apply(transform, "P06_scene_01", "P06_unit_01");
            ResolveShareTelemetry();
            // T015-B K segment 2: read the launch query once per process; a
            // friend-card open fires evt_share(stage=card_opened) and arms a
            // one-shot pending recipe. Organic opens stay inactive (zero drift).
            ScentReturn.CaptureFromPlatform(_shareTelemetry);
            // U054b-B2: the same fixed launch query feeds the gift import
            // channel; organic opens stay Absent and never reach the toast.
            GiftImport.CaptureFromPlatform(Sy.Platform.AdCapPolicy.TodayKey());
            BuildTipPanel();
            BuildSharedReturnPanel();
            BuildSeasonShareButton();
            BuildLettersButton();
            BuildAboutButton();
            BuildGiftImportToast();
        }
    }
}
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}
/// <summary>Boot-registered telemetry when the boot scene ran first
/// (BattleLauncher convention); the fallback keeps direct scene entry
/// working for tests and editor play.</summary>
void ResolveShareTelemetry()
{
    if (!ServiceLocator.TryGet<Sy.Telemetry.TelemetryService>(out _shareTelemetry))
    {
        _shareTelemetry = new Sy.Telemetry.TelemetryService();
        _shareTelemetry.AddSink(new Sy.Telemetry.DebugLogSink());
    }
}
/// <summary>T015-B: the friend's-scent return banner. Only a real
/// src=share open with a resolvable recipe shows the face - anything
/// else (organic open, unknown recipe id) shows nothing, so headless
/// gates and fresh installs stay byte-identical. The button walks the
/// player straight into the shop to craft the same scent.</summary>
void BuildSharedReturnPanel()
{
    if (!ScentReturn.OpenedFromShare || !ScentReturn.HasPendingRecipe) return;
    string nameKey = ScentCardComposer.NameKeyOf(ScentReturn.PendingRecipe);
    if (string.IsNullOrEmpty(nameKey)) return; // unknown recipe: no dead banner
    string recipeName = StoryBoard.Text(nameKey);
    if (recipeName == nameKey) return; // unresolved key: no key-name face
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
    var panel = MakePanel(canvas.transform, "SharedReturnPanel",
        new Vector2(0, 60), new Vector2(760, 340));
    MakeLabel(panel, StoryBoard.Text(ScentReturn.BannerTitleKey), 34,
        new Vector2(0, 122), new Vector2(720, 48), TipBrown);
    MakeLabel(panel, string.Format(StoryBoard.Text(ScentReturn.BannerLineKey), recipeName), 40,
        new Vector2(0, 28), new Vector2(720, 64), InkBrown);
    var go = MakeButton(panel, StoryBoard.Text(ScentReturn.BannerButtonKey),
        new Vector2(0, -102), new Vector2(360, 100));
    go.onClick.AddListener(() =>
    {
        Destroy(panel.gameObject);
        StartBusiness();
    });
}
/// <summary>T015-E: the season panorama share face (three-piece #3).
/// The shop-identity brag button only appears when a platform service
/// can carry the payload (no dead buttons) and the label key resolves
/// to real copy; the compose-at-tap ReadPixels stall rides a
/// user-initiated menu frame, never a battle frame.</summary>
void BuildSeasonShareButton()
{
    if (!ServiceLocator.TryGet<Sy.Platform.IPlatformService>(out _)) return;
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
```

```
string label = StoryBoard.Text(SeasonCardComposer.ShareButtonKey);
if (label == SeasonCardComposer.ShareButtonKey) return; // unresolved key: no key-name face
var go = MakeButton((RectTransform)canvas.transform, label,
    new Vector2(0, -780), new Vector2(420, 110));
go.onClick.AddListener(OnSeasonShareTapped);
}

/// <summary>T022: the guest mailbox entry. Letters arrive with the
/// weekly flip, so the button shows once at least one letter is
/// unlocked (fresh installs already carry two) and the label key
/// resolves - no dead face, no key-name face. The badge counts
/// unseen unlocked mail only.</summary>
void BuildLettersButton()
{
    if (LetterInbox.UnlockedLetters().Count == 0) return;
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
    string label = StoryBoard.Text(LetterPanel.KeyMailboxButton);
    if (label == LetterPanel.KeyMailboxButton) return; // unresolved key: no key-name face
    var go = MakeButton((RectTransform)canvas.transform, label,
        new Vector2(-380, -780), new Vector2(260, 110));
    go.onClick.AddListener(OpenLetters);
    _lettersBadge = MakeLabel((RectTransform)go.transform, string.Empty, 34,
        new Vector2(78, 34), new Vector2(90, 52), MilkWhite);
    RefreshLettersBadge();
}

void OpenLetters()
{
    if (_letterPanel == null)
    {
        var canvas = FindObjectOfType<Canvas>();
        if (canvas == null) return;
        _letterPanel = gameObject.AddComponent<LetterPanel>();
        _letterPanel.Initialize(canvas);
        _letterPanel.Closed += RefreshLettersBadge;
    }
    _letterPanel.Open();
}

/// <summary>T024: the about/disclosure entry. A compliance face, so
/// unlike the mailbox it always shows when the canvas and the label
/// key resolve - no service dependency, no unlock curve. The panel
/// carries the AIGC disclosure, the rare-guest odds and the ad-cap
/// lines, all composed from the live law tables.</summary>
void BuildAboutButton()
{
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
    string label = StoryBoard.Text(AboutPanel.KeyAboutButton);
    if (label == AboutPanel.KeyAboutButton) return; // unresolved key: no key-name face
    var go = MakeButton((RectTransform)canvas.transform, label,
        new Vector2(380, -780), new Vector2(260, 110));
```

```
        go.onClick.AddListener(OpenAbout);
    }
    void OpenAbout()
    {
        if (_aboutPanel == null)
        {
            var canvas = FindObjectOfType<Canvas>();
            if (canvas == null) return;
            _aboutPanel = gameObject.AddComponent<AboutPanel>();
            _aboutPanel.Initialize(canvas);
        }
        _aboutPanel.Open();
    }
    void RefreshLettersBadge()
    {
        if (_lettersBadge == null) return;
        int unread = LetterInbox.UnreadCount();
        _lettersBadge.text = unread > 0 ? unread.ToString() : string.Empty;
        _lettersBadge.gameObject.SetActive(unread > 0);
    }
    /// <summary>T015-E/T015-F funnel: compose the panorama png at tap,
    /// hand it to the platform service, and record pano_generated +
    /// pano_shared for share attribution (offline data only, red line
    /// 12). T015-F: the chat-preview face goes 1:1 - platform share
    /// thumbnails center-crop 3:4 cards and lose the title band.</summary>
    void OnSeasonShareTapped()
    {
        var spec = SeasonCardComposer.ComposeSquare();
        if (spec == null) return;
        if (!ServiceLocator.TryGet<Sy.Platform.IPlatformService>(out var platform)) return;
        var composer = new Sy.Social.ShareCardService();
        var result = composer.Compose(spec);
        if (!result.Ok)
        {
            Debug.LogWarning("[T015E] season card compose failed: " + result.FailReason);
            return;
        }
        _shareTelemetry.Track(Sy.Telemetry.EventIds.Share, new Dictionary<string, string>
        {
            { "stage", "pano_generated" },
        });
        // U054b-B1: the share link carries today's deterministic gift
        // package (gf= fragment) so a friend's open imports it and both
        // sides gain materials. Both grants land in the shared pantry
        // (PantryState) - the friend's via the B2 boot channel, the
        // sharer's once-daily reward right here. Honest absence: an
        // empty fragment leaves the query untouched.
        string dayKey = Sy.Platform.AdCapPolicy.TodayKey();
        var giftOracle = new GiftLinkOracle(PantryState.Stock);
        string giftQuery = giftOracle.ExportQuery(dayKey);
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    string query = giftQuery.Length > 0 ? spec.Query + "&" + giftQuery : spec.Query;
    platform.Share(new Sy.Platform.SharePayload
    {
        Title = spec.Title,
        ImageUrl = result.Path,
        Query = query,
    });
    _shareTelemetry.Track(Sy.Telemetry.EventIds.Share, new Dictionary<string, string>
    {
        { "stage", "pano_shared" },
    });
    // U054b-B3: the sharer's once-daily reward banks into the
    // shared pantry the caravan reads; a cap-spent or full-pantry
    // day honestly shows nothing.
    if (giftOracle.TryGrantShareReward(dayKey))
        ShowShareRewardToast(dayKey, giftOracle);
    ShowGiftCodePanel(giftOracle.BuildCode(dayKey));
}
// U054b-B1 gift-code receipt face keys (cfg_i18n, dual copy).
const string GiftCodeTitleKey = "gift_code_title";
const string GiftCodeHintKey = "gift_code_hint";
const string GiftCodeCloseKey = "gift_code_close";
/// &summary&U054b-B1: the copyable gift-code face. After a carried
/// share fires, a compact panel shows today's deterministic code so
/// the sharer can also hand-copy it (no clipboard seam - the visible
/// text is the copy). Honest absence: no panel for an empty code or
/// an unresolved key, so headless gates stay byte-identical.&/summary&
void ShowGiftCodePanel(string code)
{
    if (string.IsNullOrEmpty(code)) return;
    string title = StoryBoard.Text(GiftCodeTitleKey);
    if (title == GiftCodeTitleKey) return; // unresolved key: no key-name face
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
    var panel = MakePanel(canvas.transform, "GiftCodePanel",
        new Vector2(0, 160), new Vector2(760, 360));
    MakeLabel(panel, title, 34, new Vector2(0, 124), new Vector2(720, 48), TipBrown);
    MakeLabel(panel, code, 56, new Vector2(0, 46), new Vector2(720, 76), InkBrown);
    MakeLabel(panel, StoryBoard.Text(GiftCodeHintKey), 28,
        new Vector2(0, -34), new Vector2(700, 92), InkBrown);
    var close = MakeButton(panel, StoryBoard.Text(GiftCodeCloseKey),
        new Vector2(0, -118), new Vector2(340, 92));
    close.onClick.AddListener(() => Destroy(panel.gameObject));
}
// U054b-B3 sharer daily-reward toast key (cfg_i18n, dual copy).
const string GiftShareRewardKey = "gift_share_reward_toast";
/// &summary&U054b-B3: the sharer's daily-reward toast. Only a real
/// grant shows a face (cap spent / pantry full stay silent), the
/// line names the granted material, and an unresolved key or
/// material name means no face at all - headless gates stay

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/// byte-identical.&lt;/summary>
void ShowShareRewardToast(string dayKey, GiftLinkOracle oracle)
{
    string line = StoryBoard.Text(GiftShareRewardKey);
    if (line == GiftShareRewardKey) return; // unresolved key: no key-name face
    string matName = MaterialNameOf(oracle.ComposePackage(dayKey)[0].MaterialId);
    if (matName.Length == 0) return; // unresolved material name: no face
    var canvas = FindObjectOfType<Canvas>();
    if (canvas == null) return;
    var panel = MakePanel(canvas.transform, "ShareRewardToast",
        new Vector2(0, 430), new Vector2(640, 150));
    MakeLabel(panel, string.Format(line, matName), 30,
        new Vector2(0, 30), new Vector2(600, 76), InkBrown);
    var ok = MakeButton(panel, StoryBoard.Text(GiftCodeCloseKey),
        new Vector2(0, -42), new Vector2(280, 76));
    ok.onClick.AddListener(() => Destroy(panel.gameObject));
}

static string MaterialNameOf(int id)
{
    var materials = RecipeConfig.MaterialsBuiltIn();
    for (int i = 0; i < materials.Length; i++)
    {
        if (materials[i].Id != id) continue;
        string name = StoryBoard.Text(materials[i].NameKey);
        return name == materials[i].NameKey ? "" : name; // unresolved: empty
    }
    return "";
}

// U054b-B2 import outcome toast keys (cfg_i18n, dual copy).
const string GiftImportOkKey = "gift_import_ok_toast";
const string GiftImportDupKey = "gift_import_dup_toast";
const string GiftImportCapKey = "gift_import_cap_toast";
const string GiftImportFullKey = "gift_import_full_toast";
/// &lt;/summary>U054b-B2: the boot import toast. A gift-link open
/// names its outcome honestly - granted / duplicate / daily cap /
/// pantry full each get their own line, so the receiver always
/// knows what happened and why. Absent (organic open) and
/// Invalid (garbage link) show no face; an unresolved key shows
/// no key-name text - headless gates stay byte-identical.&lt;/summary>
void BuildGiftImportToast()
{
    string key;
    switch (GiftImport.State)
    {
        case GiftLinkOracle.ImportState.Granted: key = GiftImportOkKey; break;
        case GiftLinkOracle.ImportState.Duplicate: key = GiftImportDupKey; break;
        case GiftLinkOracle.ImportState.DailyCap: key = GiftImportCapKey; break;
        case GiftLinkOracle.ImportState.Full: key = GiftImportFullKey; break;
        default: return; // Absent / Invalid: honest silence
    }
}

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        string line = StoryBoard.Text(key);
        if (line == key) return; // unresolved key: no key-name face
        var canvas = FindObjectOfType<Canvas>();
        if (canvas == null) return;
        if (GiftImport.State == GiftLinkOracle.ImportState.Granted)
            line = string.Format(line, GiftImport.GrantedCount);
        var panel = MakePanel(canvas.transform, "GiftImportToast",
            new Vector2(0, 320), new Vector2(640, 170));
        MakeLabel(panel, line, 30, new Vector2(0, 30), new Vector2(600, 76), InkBrown);
        var ok = MakeButton(panel, StoryBoard.Text(GiftCodeCloseKey),
            new Vector2(0, -50), new Vector2(280, 80));
        ok.onClick.AddListener(() => Destroy(panel.gameObject));
    }

    /// <summary>W11b: build the offline tip panel only when tips are pending.
    /// Service lookups are guarded so opening S06_Main before boot (or in a
    /// bare harness) simply skips the panel - zero drift for headless gates.</summary>
    void BuildTipPanel()
    {
        if (!ServiceLocator.TryGet<ISaveService>(out var save)) return;
        if (!ServiceLocator.TryGet<ITimeService>(out var clock)) return;
        _clock = clock;
        _tipCounter = new TipCounter(
            IdleConfig.Formula(IdleConfig.BuiltIn(), level: 0), save, clock.ServerNowMs());
        // T012-B shelf consumption point: owned shelves raise the
        // offline tip income by TipBonusPercent (launcher-side scale,
        // the TipCounter core the settle gate pins stays untouched).
        var facility = new FacilityOracle(FacilityConfig.BuiltIn(), FacilitySave.Load());
        double tipScale = 1.0 + facility.TipBonusPercent / 100.0;
        int coins = (int)Math.Floor(_tipCounter.PeekPending(clock.ServerNowMs()) * tipScale);
        if (coins <= 0) return;
        var canvas = FindObjectOfType<Canvas>();
        if (canvas == null) return;
        var panel = MakePanel(canvas.transform, "TipPanel", new Vector2(0, -400), new Vector2(760, 300));
        MakeLabel(panel, NewbieConfig.TipPanelTitle, 34, new Vector2(0, 92), new Vector2(720, 48));
        var amount = MakeLabel(panel, string.Format(NewbieConfig.TipPanelAmount, coins), 58,
            new Vector2(0, 12), new Vector2(720, 84), InkBrown);
        var claim = MakeButton(panel, NewbieConfig.TipPanelClaim, new Vector2(0, -86), new Vector2(340, 96));
        claim.onClick.AddListener(() =>
        {
            double got = _tipCounter.Claim(_clock.ServerNowMs()) * tipScale;
            amount.text = string.Format(NewbieConfig.TipPanelClaimed, (int)Math.Floor(got));
            claim.interactable = false;
        });
    }

    static readonly Color MistPink = new Color(0.831f, 0.647f, 0.647f, 0.55f); // panel bg
    static readonly Color SageGreen = new Color(0.561f, 0.659f, 0.525f); // claim button
    static readonly Color MilkWhite = new Color(0.961f, 0.941f, 0.910f); // button label
    static readonly Color InkBrown = new Color(0.45f, 0.36f, 0.30f); // amount text
    static readonly Color TipBrown = new Color(0.722f, 0.525f, 0.420f); // title text
    static RectTransform MakePanel(Transform parent, string name, Vector2 pos, Vector2 size)

```

```

    {
        var go = new GameObject(name, typeof(RectTransform), typeof(Image));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        go.GetComponent<Image>().color = MistPink;
        return rt;
    }
    static Text MakeLabel(RectTransform parent, string text, int size, Vector2 pos, Vector2 dims, Color? color = null)
    {
        var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = dims;
        var t = go.GetComponent<Text>();
        t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        t.text = text;
        t.fontSize = size;
        t.alignment = TextAnchor.MiddleCenter;
        t.color = color ?? TipBrown;
        return t;
    }
    static Button MakeButton(RectTransform parent, string label, Vector2 pos, Vector2 size)
    {
        var go = new GameObject("ClaimButton", typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        go.GetComponent<Image>().color = SageGreen;
        var txt = MakeLabel(rt, label, 44, Vector2.zero, size);
        txt.color = MilkWhite;
        txt.raycastTarget = false;
        return go.GetComponent<Button>();
    }
}

// AboutPanel.cs
using System;
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// T024 compliance surface - the about/disclosure face on the main menu

```

```
/// (platform AIGC labeling rule + random-content disclosure). Every
/// disclosure line is composed FROM the live law tables (scent lure,
/// ad point) so the published numbers can never drift from the enforced
/// ones - the same construction guarantee the P02 odds panel uses.
/// Programmatic UGUI in the Morandi palette, zero scene-YAML drift,
/// presentation-only; the composer layer is pure static the headless
/// gate asserts without rendering.
/// </summary>
public sealed class AboutPanel : MonoBehaviour
{
    public const string KeyAboutButton = "about_button";
    public const string KeyPanelTitle = "about_title";
    public const string KeyAigcLine = "about_aigc_line";
    public const string KeyOddsLine = "about_odds_line";
    public const string KeyAdCapLine = "about_adcap_line";
    public const string KeyAgeLine = "about_age_line";
    public const string KeyCloseLabel = "about_close";
    static readonly Color SageGreen = new Color(0.561f, 0.659f, 0.525f);
    static readonly Color MilkWhite = new Color(0.961f, 0.941f, 0.910f);
    static readonly Color OchreBrown = new Color(0.722f, 0.525f, 0.420f);
    static readonly Color InkBrown = new Color(0.45f, 0.36f, 0.30f);
    // ——— composer layer (pure static, gate-facing) ———
    /// </summary>AIGC explicit disclosure: art and audio assets in this
    /// game are AI-assisted (platform AI-generated-content labeling rule;
    /// the implicit platform-side declaration stays an admin item).</summary>
    public static string AigcLine()
    {
        return StoryBoard.Text(KeyAigcLine);
    }
    /// </summary>Rare-guest odds disclosure. The published numbers are
    /// formatted straight out of the live scent-lure table, so the shown
    /// face and the enforced law are the same object - a hot-tuned table
    /// re-renders the disclosure with zero extra wiring.</summary>
    public static string OddsLine()
    {
        var law = ScentLureConfig.BuiltIn();
        return string.Format(StoryBoard.Text(KeyOddsLine),
            Percent(law.RareBaseChance), Percent(law.RareWithAdChance), law.RarePityAt);
    }
    /// </summary>Ad-cap disclosure: the global daily ceiling formatted out
    /// of the live ad-point law (voluntary views only - no forced ad).</summary>
    public static string AdCapLine()
    {
        var law = AdPointConfig.BuiltIn();
        return string.Format(StoryBoard.Text(KeyAdCapLine), law.GlobalCapDaily);
    }
    /// </summary>Age-rating disclosure (IAA casual default: 8+).</summary>
    public static string AgeLine()
    {
        return StoryBoard.Text(KeyAgeLine);
    }
}
```

```
}
static int Percent(double chance)
{
    return (int)Math.Round(chance * 100.0);
}
// —— view layer ——
RectTransform _overlay;
public void Initialize(Canvas canvas)
{
    _overlay = new GameObject("AboutOverlay", typeof(RectTransform), typeof(Image)).GetComponent<RectTransform>();
    _overlay.transform.SetParent(canvas.transform, false);
    _overlay.anchorMin = Vector2.zero;
    _overlay.anchorMax = Vector2.one;
    _overlay.offsetMin = _overlay.offsetMax = Vector2.zero;
    var dim = _overlay.GetComponent<Image>().sprite;
    dim.sprite = MakeSquareSpriteShared();
    dim.color = new Color(0.35f, 0.30f, 0.27f, 0.92f);
    var paper = MakeCard(_overlay, "AboutPaper", Vector2.zero, new Vector2(940, 1180),
        new Color(MilkWhite.r, MilkWhite.g, MilkWhite.b, 0.97f));
    MakeText(paper, StoryBoard.Text(KeyPanelTitle), 64, new Vector2(0, 460),
        TextAnchor.MiddleCenter, 900, 110).color = OchreBrown;
    MakeText(paper, AigcLine(), 36, new Vector2(0, 250),
        TextAnchor.MiddleCenter, 820, 120).color = InkBrown;
    MakeText(paper, OddsLine(), 34, new Vector2(0, 60),
        TextAnchor.MiddleCenter, 820, 190).color = InkBrown;
    MakeText(paper, AdCapLine(), 34, new Vector2(0, -130),
        TextAnchor.MiddleCenter, 820, 110).color = InkBrown;
    MakeText(paper, AgeLine(), 34, new Vector2(0, -250),
        TextAnchor.MiddleCenter, 820, 110).color = InkBrown;
    MakeButton(paper, StoryBoard.Text(KeyCloseLabel), new Vector2(0, -440),
        new Vector2(420, 96), Close);
    _overlay.gameObject.SetActive(false);
}
public void Open()
{
    _overlay.gameObject.SetActive(true);
}
void Close()
{
    _overlay.gameObject.SetActive(false);
}
// —— tools (programmatic UGUI, zero scene YAML) ——
static RectTransform MakeCard(RectTransform parent, string name, Vector2 pos, Vector2 size, Color color)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = size;
```

```
        var img = go.GetComponent<Image>();
        img.sprite = MakeSquareSpriteShared();
        img.color = color;
        return rt;
    }
    static Text MakeText(RectTransform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
    {
        var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(w, h);
        var t = go.GetComponent<Text>();
        t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        t.text = text;
        t.fontSize = size;
        t.alignment = anchor;
        t.color = InkBrown;
        return t;
    }
    static void MakeButton(RectTransform parent, string label, Vector2 pos, Vector2 size, Action onClick)
    {
        var go = new GameObject("ActionButton", typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        var img = go.GetComponent<Image>();
        img.sprite = MakeSquareSpriteShared();
        img.color = SageGreen;
        var txt = MakeText(rt, label, 40, Vector2.zero, TextAnchor.MiddleCenter, size.x, size.y);
        txt.color = MilkWhite;
        txt.raycastTarget = false;
        go.GetComponent<Button>().onClick.AddListener(() => onClick());
    }
    static Sprite _sharedSquare;
    static Sprite MakeSquareSpriteShared()
    {
        if (_sharedSquare == null)
        {
            var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
            var px = tex.GetPixels32();
            for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
            tex.SetPixels32(px);
            tex.Apply(false, false);
            _sharedSquare = Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
        }
        return _sharedSquare;
    }
```

```
    }
}
}
// AdPointTable.cs
using System;
using System.Collections.Generic;
namespace Sy.P06.Newbie
{
    /// <summary>Ad point kind: nine rewarded video points plus the single
    /// interstitial face (shop-rest settle only, the most restrained slot in
    /// the design table).</summary>
    public enum AdPointType { Rewarded, Interstitial }
    /// <summary>One ad point row (pitch doc section six, V1.2 values).</summary>
    public readonly struct AdPointEntry
    {
        public readonly string Id;
        public readonly string Category;
        public readonly string Surface;
        public readonly int CapPerDay;
        public readonly AdPointType Type;
        public readonly int Priority;
        public AdPointEntry(string id, string category, string surface,
            int capPerDay, AdPointType type, int priority)
        {
            Id = id;
            Category = category;
            Surface = surface;
            CapPerDay = capPerDay;
            Type = type;
            Priority = priority;
        }
    }
}
/// <summary>Flat cfg_ad_point values (authoring table:
/// Design/configs/P06/cfg_ad_point.json).</summary>
public readonly struct AdPointLaw
{
    public readonly int Version;
    public readonly bool Disclosed;
    public readonly int GlobalCapDaily;
    public readonly int InterstitialCapDaily;
    public readonly double InterstitialCooldownSec;
    public readonly AdPointEntry[] Points;
    public AdPointLaw(int version, bool disclosed, int globalCapDaily,
        int interstitialCapDaily, double interstitialCooldownSec, AdPointEntry[] points)
    {
        Version = version;
        Disclosed = disclosed;
        GlobalCapDaily = globalCapDaily;
        InterstitialCapDaily = interstitialCapDaily;
        InterstitialCooldownSec = interstitialCooldownSec;
    }
}
```

```

        Points = points;
    }
}
/// <summary>
/// T014 ad point table (task book: 10 ad points = 9 rewarded + 1
/// interstitial, pitch doc section six, V1.2). Mirrors the scent-lure/
/// caravan/codex/facility table pattern: BuiltIn() is the embedded slice
/// the live game runs on while Design/configs/P06/cfg_ad_point.json is the
/// authoring table the ad-point gate validates against (drift guard: the
/// two must agree row for row). Every rewarded row owns its category so
/// the shared AdCapPolicy enforces the per-point cap on top of the 12/day
/// global cap; the interstitial row carries its own 3/day cap and the
/// 180s cooldown on the policy's interstitial face. Priority is a
/// declared ordering for surfaces and analytics - the global cap stays
/// first-come-first-served in v1.0, which the schema desc states openly.
/// Id namespaces stay isolated from P02's funnel: the idle-earnings point
/// is "idle_double", never P02's "offline_double", so a P06 call can
/// never draw the P02 double budget (same isolation law the supply
/// category already documents).
/// </summary>
public static class AdPointConfig
{
    static readonly AdPointEntry[] Embedded =
    {
        new AdPointEntry("idle_double", "idle_double", "harvest", 3, AdPointType.Rewarded, 50),
        new AdPointEntry("rare_guest", "rare_guest", "shop", 3, AdPointType.Rewarded, 60),
        new AdPointEntry("craft_speed", "craft_speed", "craft", 3, AdPointType.Rewarded, 40),
        new AdPointEntry("caravan_supply", "supply", "caravan", 3, AdPointType.Rewarded, 40),
        new AdPointEntry("recipe_hint", "recipe_hint", "craft", 1, AdPointType.Rewarded, 55),
        new AdPointEntry("sign_double", "sign_double", "signin", 1, AdPointType.Rewarded, 30),
        new AdPointEntry("expand_speed", "expand_speed", "expand", 1, AdPointType.Rewarded, 45),
        new AdPointEntry("story_preview", "story_preview", "story", 1, AdPointType.Rewarded, 20),
        new AdPointEntry("book_complete", "book_complete", "codex", 2, AdPointType.Rewarded, 25),
        new AdPointEntry("interstitial", "interstitial", "settle", 3, AdPointType.Interstitial, 10),
    };
    static readonly AdPointLaw EmbeddedLaw = new AdPointLaw(
        1, true, 12, 3, 180.0, Embedded);
    public static AdPointLaw BuiltIn() => EmbeddedLaw;
    /// <summary>Parse cfg_ad_point.json content (MiniJson: long/double).
    /// A malformed root returns false so callers fall back to the
    /// embedded table; missing fields parse as zero/empty so the
    /// zero-drift gate catches authoring gaps.</summary>
    public static bool TryFromJson(string json, out AdPointLaw law)
    {
        law = default;
        var root = Sy.Core.MiniJson.Deserialize(json) as Dictionary<string, object>;
        if (root == null) return false;
        AdPointEntry[] points = null;
        if (root.TryGetValue("points", out var rawRows) && rawRows is List<object> rows)
        {

```

```

        points = new AdPointEntry[rows.Count];
        for (int i = 0; i < rows.Count; i++)
        {
            if (!(rows[i] is Dictionary<string, object> row)) continue;
            points[i] = new AdPointEntry(
                AsString(row, "id"),
                AsString(row, "category"),
                AsString(row, "surface"),
                AsInt(row, "capPerDay"),
                ParseType(AsString(row, "type")),
                AsInt(row, "priority"));
        }
    }
    law = new AdPointLaw(
        AsInt(root, "version"),
        AsBool(root, "disclosed"),
        AsInt(root, "globalCapDaily"),
        AsInt(root, "interstitialCapDaily"),
        AsDouble(root, "interstitialCooldownSec"),
        points);
    return true;
}

static AdPointType ParseType(string raw)
    => raw == "interstitial" ? AdPointType.Interstitial : AdPointType.Rewarded;
static int AsInt(Dictionary<string, object> row, string key)
    => row.TryGetValue(key, out var v) && v is long l ? (int)l : 0;
static double AsDouble(Dictionary<string, object> row, string key)
{
    if (!row.TryGetValue(key, out var v)) return 0;
    if (v is double d) return d;
    if (v is long l) return l;    // MiniJson yields long for integer literals
    return 0;
}

static bool AsBool(Dictionary<string, object> row, string key)
    => row.TryGetValue(key, out var v) && v is bool b && b;
static string AsString(Dictionary<string, object> row, string key)
    => row.TryGetValue(key, out var v) && v is string s ? s : "";
}

}

// BattleLauncher.cs
using System.Collections.Generic;
using UnityEngine;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// S06_Battle      ShopSim+TutorialFlow+ShopView    Tick
    ///   Sy.Telemetry Editor =Debug sink T001e          +TCB
    /// HUD              6
    /// </summary>
    public sealed class BattleLauncher : MonoBehaviour

```

```

{
    ShopSim _sim;
    TutorialFlow _tutorial;
    ShopView _view;
    StoryPanel _storyPanel; // T009: guest-chapter story panel (weekly flip)
    CaravanOracle _caravan; // T011: caravan supply loop (restock cycle)
    CaravanPanel _caravanPanel; // T011: restock sheet (missing-material popup)
    CodexOracle _codex; // T013: dual-codex progress (guests + recipes)
    CodexPanel _codexPanel; // T013-B: the dual-codex page (tier reward claim)
    FacilityOracle _facility; // T012: expansion engine (owned set from local save)
    FacilityPanel _facilityPanel; // T012-B: the store expansion page (purchase surface)
    RecipeHintOracle _hintOracle; // T014-B: recipe_hint point law (formal craft face)
    CraftSpeedOracle _speedOracle; // T014-B: craft_speed point law (formal brewing face)
    StoryPreviewOracle _previewOracle; // T014-B: story_preview point law (locked-chapter face)
    BookCompleteOracle _bookOracle; // T014-B: book_complete point law (codex claim doubler face)
    RareGuestBoostOracle _rareBoostOracle; // T014-B: rare_guest point law (VIP-lure boost face)
    SignDoubleOracle _signOracle; // T014-B: sign_double point law (daily sign-in doubler face)
    InterstitialSettleOracle _interstitial; // T014-B: interstitial point law (shop-rest settle face)
    bool _formalSession; // formal-loop flag: the tutorial stays ad-free (zero-ad red line before M5)
    SignInPanel _signPanel; // T014-B: the daily sign-in sheet (D1-D7 streak surface)
    Sy.Idle.SignInSystem _signIn; // T014-B: the claim law over the reward ladder
    bool _rareBoostArmed; // T014-B: the next lure rolls at the boosted rare chance
    bool _lastLureReady; // T014-B: latch for pushing lure-ready changes to the view
    Sy.Telemetry.TelemetryService _telemetry;
    Sy.Social.ShareCard _scentSpec; // T015-A: 3:4 scent-card share face
    Sy.Social.ComposeResult _scentCard;
    Sy.Platform.IAlbum _album; // T015-C: save-to-album seam (null = no bridge, save face stays hidden)
    ScentAtmosphere _atmosphere;
    RareGuestOracle _rareOracle; // T010 rare-guest law rider on the scent lure
    ReceptionOracle _reception; // T002 reception-seat law (formal loop; tutorial stays inert)
    ScentLureValues _lureLaw;
    System.Random _rng;
    P06AudioHost _audioHost; // X1675 matrix#44 slice-b: five-beat SFX host
    int _lastCoins = -1, _lastHearts = -1;
    void Awake()
    {
        Application.targetFrameRate = 30;
        var canvas = FindObjectOfType<Canvas>();
        _sim = new ShopSim();
        _view = gameObject.AddComponent<ShopView>();
        _view.Initialize(canvas, _sim);
    }
}

#if DEVELOPMENT_BUILD || UNITY_EDITOR
    // U133 / doc 12 section 6: dev-only one-tap bug report. P06 has no
    // seed system - the context block carries the live shop state.
    Sy.Quality.BugReportService.Game = "P06";
    Sy.Quality.BugReportService.Version = Application.version;
    Sy.Quality.BugReportService.SeedInfo = "";
    Sy.Quality.BugReportService.ContextJson = () => P06QualityContext(_sim);
    Sy.Quality.BugReportButton.Attach(canvas.transform);
#endif

```

```
// T009: story panel rides the same canvas; the HUD story
// button is its only entry point (presentation-only).
_storyPanel = gameObject.AddComponent<StoryPanel>();
_storyPanel.Initialize(canvas);
// T011: caravan supply loop + restock sheet; the HUD caravan
// button opens it. The oracle owns the shared AdCapPolicy
// consume (P06's mock ad service has no cap logic - see the
// oracle doc for the T001c suite split). U054b-B3: the oracle
// reads the shared pantry (PantryState) so gift imports and
// share rewards land in the stock this sheet displays.
_caravan = new CaravanOracle(CaravanConfig.BuiltIn(), PantryState.Stock);
_caravanPanel = gameObject.AddComponent<CaravanPanel>();
_caravanPanel.Initialize(canvas, _caravan);
// T013: the dual codex fills as guests arrive and recipes are
// crafted (the tutorial order is the fox scholar + rain-green-hill,
// see NewbieConfig wiring ids). The panel is the claim surface; the
// launcher owns the wallet deposit - the panel never grants.
_codex = new CodexOracle(CodexConfig.BuiltIn());
_codexPanel = gameObject.AddComponent<CodexPanel>();
_codexPanel.Initialize(canvas, _codex);
// T012-B: the expansion engine boots from the local save (red
// line 12: PlayerPrefs authority) and the panel is its purchase
// surface - the launcher owns the wallet spend and the save write.
_facility = new FacilityOracle(FacilityConfig.BuiltIn(), FacilitySave.Load());
_facilityPanel = gameObject.AddComponent<FacilityPanel>();
_facilityPanel.Initialize(canvas, _facility);
// T014-B slice 1: the recipe_hint ad point (V1.2: 1/day, craft
// surface). The oracle owns the AdCapPolicy consume (mock ad
// service has no cap logic); the tutorial keeps the hint face
// hidden (zero-ad red line before M5) - the formal craft batch
// lights it via SetHintFace(true).
_hintOracle = new RecipeHintOracle(AdPointConfig.BuiltIn());
_view.HintTapped += OnHintTapped;
_view.SetHintFace(false);
// T014-B slice 2: the craft_speed ad point (V1.2: 3/day, brewing
// wait surface). The oracle owns the AdCapPolicy consume (mock ad
// service has no cap logic); the grant skips the brewing wait so
// the recipe card flips right away. The tutorial keeps the speed
// face hidden (zero-ad red line before M5) - the formal craft
// batch lights it via SetSpeedFace(true).
_speedOracle = new CraftSpeedOracle(AdPointConfig.BuiltIn());
_view.SpeedTapped += OnSpeedTapped;
_view.SetSpeedFace(false);
// T014-B slice 3: the story_preview ad point (V1.2: 1/day,
// locked-chapter face). The oracle owns the AdCapPolicy consume
// (mock ad service has no cap logic); the grant opens the locked
// chapter's opening as a read-only preview (clues stay sealed).
// The tutorial keeps the preview face hidden (zero-ad red line
// before M5) - the formal story batch lights it via
// SetPreviewFace(true).
```

```
_previewOracle = new StoryPreviewOracle(AdPointConfig.BuiltIn());
_storyPanel.PreviewTapped += OnPreviewTapped;
_storyPanel.SetPreviewFace(false);
// T015-D: the chapter-detail keepsake share face. Sharing is not
// an ad point, so the face lights whenever a platform service can
// carry the payload (no dead buttons) - the tutorial may share too.
_storyPanel.StoryShareTapped += OnStoryShareTapped;
_storyPanel.SetStoryShareFace(Sy.Core.ServiceLocator.TryGet<Sy.Platform.IPlatformService>(out _));
// T014-B slice 4: the book_complete ad point (V1.2: 2/day, the
// codex tier-claim doubler face). The oracle owns the AdCapPolicy
// consume (mock ad service has no cap logic); the grant doubles
// the claim it rides on. The tutorial keeps the doubler face
// hidden (zero-ad red line before M5) - the formal codex batch
// lights it via SetDoubleFace(true).
_bookOracle = new BookCompleteOracle(AdPointConfig.BuiltIn());
_codexPanel.DoubleClaimRequested += OnCodexDoubleRequested;
_codexPanel.SetDoubleFace(false);
// T014-B slice 5: the rare_guest ad point (V1.2: 3/day,
// VIP-lure-ready street face). The oracle owns the AdCapPolicy
// consume (mock ad service has no cap logic); the grant arms the
// next scent lure to roll the rare arrival at the boosted 0.8
// chance instead of the 0.05 base. The tutorial keeps the boost
// face hidden (zero-ad red line before M5) - the formal street
// batch lights it via SetRareFace(true).
_rareBoostOracle = new RareGuestBoostOracle(AdPointConfig.BuiltIn());
_view.RareTapped += OnRareBoostTapped;
_view.SetRareFace(false);
// T014-B slice 6: the sign_double ad point (V1.2: 1/day, the
// daily sign-in doubler face). The sign-in sheet opens from the
// battle HUD - the main screen stays ad-free by design - and the
// oracle owns the AdCapPolicy consume (mock ad service has no
// cap logic); the grant doubles the sign-in reward deposit. The
// tutorial keeps the doubler face hidden (zero-ad red line
// before M5) - the formal sign-in batch lights it via
// SetDoubleFace(true).
_signOracle = new SignDoubleOracle(AdPointConfig.BuiltIn());
_signIn = new Sy.Idle.SignInSystem(NewbieConfig.SignInRewardTable);
_signPanel = gameObject.AddComponent<SignInPanel>();
_signPanel.Initialize(canvas);
_signPanel.SetDoubleFace(false);
// T014-B slice 7: the interstitial point (V1.2: 3/day + 180s
// cooldown, shop-rest settle face). No button, no reward - the
// platform suite owns the cooldown + cap consume, so the oracle
// stays a thin formal-session gate.
_interstitial = new InterstitialSettleOracle(AdPointConfig.BuiltIn());
// T014-B formal-loop slice: the settle "reopen" reloads the
// battle scene, so the durable session count (PlayerPrefs,
// red line 12) is the only state that survives the reload.
// Session 0 is the tutorial - every ad face stays dark on that
// boot (zero-ad red line before M5); every later boot is a
```

```
// formal session and lights all six faces at once (each one
// still self-gates on its own live moment via SyncView).
_formalSession = FormalSessionSave.Load() &gt;= 1;
if (_formalSession)
{
    _view.SetHintFace(true);
    _view.SetSpeedFace(true);
    _storyPanel.SetPreviewFace(true);
    _codexPanel.SetDoubleFace(true);
    _view.SetRareFace(true);
    _signPanel.SetDoubleFace(true);
}
// T006 scent atmosphere: orders fill the street, the VIP follows the scent.
_atmosphere = new ScentAtmosphere();
_view.SetScentSource(_atmosphere);
_atmosphere.VipLureTriggered += OnVipLureTriggered;
// T002 reception slice: the formal loop seats every lured VIP
// (rare guests reuse their stage seconds as the stay) and pays
// a departure tip scaled by the shelf family; a full house
// turns the guest away. The tutorial keeps the oracle inert so
// the pinned settle math never moves - capacity reads the
// facility engine live, so a mid-run expansion is effective
// on the very next lure.
_reception = new ReceptionOracle(
    _formalSession,
    () => _facility.GuestCapacity,
    () => _facility.TipBonusPercent);
_view.SetSeatedFace(0, _reception.Capacity);
// T010 rare-guest law (embedded slice; VipRareCheck guards zero
// drift against Design/configs/P06/cfg_scent_lure.json).
_lureLaw = ScentLureConfig.BuiltIn();
_rareOracle = new RareGuestOracle(_lureLaw);
_rng = new System.Random();
// X1675 matrix#44 slice-b: scene-local audio host with the five
// teaching-beat SFX wiring (accept click / invest feedback triad
// / craft success / codex reveal / settle tips). The launcher
// owns this mapping (P02 X813 paradigm) so the editor census
// mirrors the exact production path through the probe ledger.
// Null-safe law: a missing clip stays silent in any build state.
_audioHost = P06AudioHost.Ensure("bgm_shop", 0.6f);
_sim.OrderAccepted += () => _audioHost.PlaySfx("sfx_ui_click", 0.8f);
_sim.FeedbackGiven += (slot, materialId, state, trueSlot)
    => _audioHost.PlaySfx(FeedbackSfxKey(state), 0.85f);
_sim.RecipeCardShown += () => _audioHost.PlaySfx("sfx_craft_success", 0.9f);
_sim.CodexUnlocked += () => _audioHost.PlaySfx("sfx_codex", 0.9f);
_sim.Settled += () => _audioHost.PlaySfx("sfx_tips", 0.9f);
_telemetry = new Sy.Telemetry.TelemetryService();
_telemetry.AddSink(new TelemetryDebugSink());
_tutorial = new TutorialFlow(
    _sim,
```

```

        hint = >
    {
        Debug.Log("[HINT] " + hint);
        _view.SetHint(hint);
    },
    step = > _telemetry.Track(Sy.Telemetry.EventIds.NewbieStep,
        new Dictionary<string, string> { { "step", step.ToString() } }));
    _view.AcceptRequested += OnAcceptRequested;
    _view.MaterialTapped += OnMaterialTapped;
    _view.SocialTapped += OnSocialTapped;
    _view.StoryTapped += OnStoryTapped;
    _view.CaravanTapped += OnCaravanTapped;
    _view.CodexTapped += OnCodexTapped;
    _view.FacilityTapped += OnFacilityTapped;
    _view.SignTapped += OnSignTapped;
    _caravanPanel.SupplyRequested += OnCaravanSupplyRequested;
    _codexPanel.ClaimRequested += OnCodexClaimRequested;
    _facilityPanel.PurchaseRequested += OnFacilityPurchaseRequested;
    _signPanel.SignInRequested += OnSignInClaimRequested;
    _signPanel.DoubleSignInRequested += OnSignInDoubleRequested;
    _sim.StateChanged += OnStateChanged;
    _sim.FeedbackGiven += OnFeedbackGiven;
    _sim.OrderCompleted += OnOrderCompleted;
    _sim.StarGained += OnStarGained;
    _sim.RecipeCardShown += OnRecipeCardShown;
    _view.ShareRequested += OnCardShareRequested; // T015-A: card-face share funnel
    _view.SaveRequested += OnCardSaveRequested; // T015-C: card-face save-to-album funnel
    _view.AlbumRetryTapped += OnAlbumRetryTapped; // T015-C: denied-save retry funnel
}

#if DEVELOPMENT_BUILD || UNITY_EDITOR
/// <summary>U133 / doc 12 section 6: dev-only report context, fresh at tap
/// time (BR-*.json context block = live shop state for reproduction).</summary>
static string P06QualityContext(ShopSim sim)
{
    if (sim == null) return "{}";
    return "{\\"coins\":" + sim.Coins + ",\\"hearts\":" + sim.Hearts
        + ",\\"stars\":" + sim.Stars + ",\\"activeSlot\":" + sim.ActiveSlot
        + ",\\"stateTimer\":" + sim.StateTimer.ToString("F1", System.Globalization.CultureInfo.InvariantCulture)
        + ",\\"correctInvests\":" + sim.CorrectInvests + "}";
}
#endif

void Update()
{
    _sim.Tick(Time.deltaTime);
    _atmosphere.Tick(Time.deltaTime);
    _view.SyncView(_sim);
    // T002 reception: advance every seated guest - departures pay
    // their tip into the wallet and pop on the HUD (formal loop
    // only; the tutorial oracle never seats or departs anyone).
    var departures = _reception.Tick(Time.deltaTime);

```

```
if (departures.Count > 0)
{
    int tipTotal = 0;
    for (int i = 0; i < departures.Count; i++) tipTotal += departures[i].TipCoins;
    _sim.Deposit(tipTotal, 0);
    _view.PlayIncomePop(tipTotal, 0);
    _view.SetSeatedFace(_reception.SeatedCount, _reception.Capacity);
    // T019 portrait pair: the seat clears when the house empties
    // (the counter HUD keeps carrying the live seat truth).
    if (_reception.SeatedCount == 0) _view.HideSeatedGuest();
}
if (_sim.Coins != _lastCoins) { _lastCoins = _sim.Coins; _view.SetCoins(_lastCoins); }
if (_sim.Hearts != _lastHearts) { _lastHearts = _sim.Hearts; _view.SetHearts(_lastHearts); }
// T014-B: push lure-ready changes to the rare face so the boost
// offer only surfaces on the live street moment.
bool lureReady = _atmosphere.VipLureReady;
if (lureReady != _lastLureReady) { _lastLureReady = lureReady; _view.SetRareFaceLive(lureReady); }
}
void OnAcceptRequested()
{
    _sim.AcceptOrder();
}
void OnMaterialTapped(int materialId)
{
    _sim.TryInvest(materialId);
}
void OnSocialTapped()
{
    _view.SetHint(" — ");
}
void OnStoryTapped()
{
    // X1680 audio face: UI tap group - panel-open click beat.
    _audioHost.PlaySfx("sfx_ui_click", 0.8f);
    _storyPanel.Open();
}
void OnCodexTapped()
{
    _audioHost.PlaySfx("sfx_ui_click", 0.8f);
    _codexPanel.Open();
}
void OnFacilityTapped()
{
    _audioHost.PlaySfx("sfx_ui_click", 0.8f);
    _facilityPanel.Open(_sim.Coins, _sim.Hearts);
}
void OnFacilityPurchaseRequested(int facilityId)
{
    // The oracle owns the affordability law; the wallet spends only
    // after a successful purchase, so coins never drain without a
```

```
// row joining the owned set (and an owned row never re-charges).
if (!_facility.TryPurchase(facilityId, _sim.Coins, _sim.Hearts, out _, out _))
{
    _facilityPanel.Refresh(_sim.Coins, _sim.Hearts);
    return;
}
var entry = _facility.Entry(facilityId);
_sim.TrySpend(entry.CostCoins, entry.CostHearts);
FacilitySave.Save(_facility.OwnedIds());
_facilityPanel.Refresh(_sim.Coins, _sim.Hearts);
}

void OnCodexClaimRequested(CodexSide side)
{
    // The ledger grants once per tier step; the deposit lands in the
    // shop wallet and the HUD picks the diff up on its next sync.
    if (_codex.TryClaimTierReward(side, out int coins, out int hearts))
    {
        _sim.Deposit(coins, hearts);
        _codexPanel.Refresh();
    }
}

void OnCaravanTapped()
{
    // X1680 audio face: UI tap group - panel-open click beat.
    _audioHost.PlaySfx("sfx_ui_click", 0.8f);
    _caravanPanel.Open();
    if (_caravan.MissingCount(_caravan.Law.ShowcaseRecipeId) > 0)
        TrackCaravan("offer");
}

async void OnCaravanSupplyRequested()
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
    if (!_caravan.CanOffer(day)) return;
    if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
    {
        var result = await ads.ShowRewarded(_caravan.Law.PointId);
        if (!result.Completed)
        {
            TrackCaravan("ad_fail");
            return;
        }
    }
    // The oracle consumes the supply-category slot itself (P06 mock
    // suite has no cap logic; T001c moves the consume into the suite).
    if (_caravan.TryCompleteSupply(day))
    {
        _caravanPanel.Refresh();
        TrackCaravan("accept");
    }
    else
}
```

```
{
    TrackCaravan("blocked");
}
}
void TrackCaravan(string state)
{
    _telemetry.Track(Sy.Telemetry.EventIds.Ad, new Dictionary<string, string>
    {
        { "state", state },
        { "pointId", _caravan.Law.PointId },
        { "reason", "caravan_supply" },
        { "missing", _caravan.MissingCount(_caravan.Law.ShowcaseRecipeId).ToString() },
    });
}
// —— T014-B recipe_hint consumption (formal craft face) ——
async void OnHintTapped()
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
    bool crafting = _sim.State == ShopSim.ShopState.Crafting;
    if (!_hintOracle.CanOffer(day, crafting, _sim.ActiveSlot))
    {
        TrackHint("blocked");
        return;
    }
    TrackHint("offer");
    if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
    {
        var result = await ads.ShowRewarded(_hintOracle.PointId);
        if (!result.Completed)
        {
            TrackHint("ad_fail");
            return;
        }
    }
    // The oracle consumes the recipe_hint slot itself (mock ad
    // service has no cap logic - CaravanOracle integration debt).
    if (_hintOracle.TryGrant(day, crafting, _sim.ActiveSlot, out int slot))
    {
        _view.ShowHint(slot);
        TrackHint("accept");
    }
    else
    {
        TrackHint("blocked");
    }
}
void TrackHint(string state)
{
    _telemetry.Track(Sy.Telemetry.EventIds.Ad, new Dictionary<string, string>
    {
```

```
        { "state", state },
        { "pointId", _hintOracle.PointId },
        { "reason", "recipe_hint" },
    });
}
// —— T014-B craft_speed consumption (formal brewing face) ——
async void OnSpeedTapped()
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
    bool brewing = _sim.State == ShopSim.ShopState.Brewing;
    if (!_speedOracle.CanOffer(day, brewing))
    {
        TrackSpeed("blocked");
        return;
    }
    TrackSpeed("offer");
    if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
    {
        var result = await ads.ShowRewarded(_speedOracle.PointId);
        if (!result.Completed)
        {
            TrackSpeed("ad_fail");
            return;
        }
    }
    // The oracle consumes the craft_speed slot itself (mock ad
    // service has no cap logic - CaravanOracle integration debt);
    // the grant lands as a brewing-wait skip, so the card flips now.
    if (_speedOracle.TryGrant(day, brewing) && _sim.TrySkipBrew())
    {
        TrackSpeed("accept");
    }
    else
    {
        TrackSpeed("blocked");
    }
}
void TrackSpeed(string state)
{
    _telemetry.Track(Sy.Telemetry.EventIds.Ad, new Dictionary<string, string>
    {
        { "state", state },
        { "pointId", _speedOracle.PointId },
        { "reason", "craft_speed" },
    });
}
// —— T014-B story_preview consumption (locked-chapter face) ——
async void OnPreviewTapped(ChapterEntry chapter)
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
```

```
// Face law: only a still-locked chapter can preview - live rows
// open free, so a flipped week never burns ad budget.
bool locked = !Storyboard.IsLive(chapter);
if (!_previewOracle.CanOffer(day, locked))
{
    TrackPreview("blocked");
    return;
}
TrackPreview("offer");
if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
{
    var result = await ads.ShowRewarded(_previewOracle.PointId);
    if (!result.Completed)
    {
        TrackPreview("ad_fail");
        return;
    }
}
// The oracle consumes the story_preview slot itself (mock ad
// service has no cap logic - CaravanOracle integration debt);
// the grant renders the read-only preview, clues stay sealed.
if (_previewOracle.TryGrant(day, locked))
{
    _storyPanel.ShowPreview(chapter);
    TrackPreview("accept");
}
else
{
    TrackPreview("blocked");
}
}

void TrackPreview(string state)
{
    _telemetry.Track(Sy.Telemetry.EventIds.Ad, new Dictionary<string, string>
    {
        { "state", state },
        { "pointId", _previewOracle.PointId },
        { "reason", "story_preview" },
    });
}

// —— T014-B book_complete consumption (codex claim doubler face) ——
async void OnCodexDoubleRequested(CodexSide side)
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
    // Face law: only a still-pending tier claim gates the offer -
    // a claimed-out book never burns ad budget.
    if (!_bookOracle.CanOffer(day, _codex.HasPendingTierReward(side)))
    {
        TrackBookDouble("blocked");
        return;
    }
}
```

```
}
TrackBookDouble("offer");
if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
{
    var result = await ads.ShowRewarded(_bookOracle.PointId);
    if (!result.Completed)
    {
        TrackBookDouble("ad_fail");
        return;
    }
}
// Re-check the face after the await: a plain claim during the ad
// flips the pending state off - the oracle then denies without
// consuming, so budget never burns without a claim to double.
bool stillPending = _codex.HasPendingTierReward(side);
if (_bookOracle.TryGrant(day, stillPending)
    && _codex.TryClaimTierReward(side, out int coins, out int hearts))
{
    _sim.Deposit(coins * 2, hearts * 2);
    _codexPanel.Refresh();
    TrackBookDouble("accept");
}
else
{
    TrackBookDouble("blocked");
}
}
void TrackBookDouble(string state)
{
    _telemetry.Track(Sy.Telemetry.EventIds.Ad, new Dictionary<string, string>
    {
        { "state", state },
        { "pointId", _bookOracle.PointId },
        { "reason", "book_complete" },
    });
}
async void OnRareBoostTapped()
{
    string day = Sy.Platform.AdCapPolicy.TodayKey();
    // Face law: only a lure-ready street gates the offer - a cold
    // street never burns ad budget.
    if (!_rareBoostOracle.CanOffer(day, _atmosphere.VipLureReady))
    {
        TrackRareBoost("blocked");
        return;
    }
    TrackRareBoost("offer");
    if (Sy.Core.ServiceLocator.TryGet<Sy.Platform.IAdsService>(out var ads))
    {
        var result = await ads.ShowRewarded(_rareBoostOracle.PointId);
```

```
        streetImgs[i].color = new Color(SageGreen.r, SageGreen.g, SageGreen.b, 0f);
        phase[i] = i * 0.7f;
    }
    float mistClock = 0f;
    while (true)
    {
        mistClock += Time.deltaTime;
        float conc = _atmosphere != null ? Mathf.Clamp01(_atmosphere.Concentration) : 0f;
        if (_lureBurst > 0f) _lureBurst -= Time.deltaTime;
        bool surge = _lureBurst > 0f;
        float doorAlpha = Mathf.Min(0.55f + conc * 0.25f + (surge ? 0.2f : 0f), 1f);
        // 8 fps 3-frame loop, staggered per puff for a rolling mist
        if (hasMist)
        {
            int mf = (int)(mistClock * 8f) % 3;
            for (int i = 0; i < puffs.Length; i++)
            {
                var f = mistFrames[(mf + i) % 3];
                if (doorImgs[i].sprite != f) doorImgs[i].sprite = f;
            }
            for (int i = 0; i < street.Length; i++)
            {
                var f = mistFrames[(mf + i) % 3];
                if (streetImgs[i].sprite != f) streetImgs[i].sprite = f;
            }
        }
        // door puffs: ambient drift up, denser while the scent lingers
        for (int i = 0; i < puffs.Length; i++)
        {
            var pos = puffs[i].anchoredPosition;
            pos.y += (55f + conc * 26f) * Time.deltaTime;
            pos.x += (i % 2 == 0 ? 14f : -10f) * Time.deltaTime;
            puffs[i].anchoredPosition = pos;
            if (pos.y > 350f) ResetPuff(puffs[i], 0f);
            var dc = doorImgs[i].color;
            dc.a = doorAlpha;
            doorImgs[i].color = dc;
        }
        // street puffs: flow out of the door toward the street, scaled by concentration
        for (int i = 0; i < street.Length; i++)
        {
            var sc = streetImgs[i].color;
            var streetBase = _rareSurge ? WarmAmber : (hasMist ? Color.white : SageGreen); // T010: amber stre
            sc.r = streetBase.r; sc.g = streetBase.g; sc.b = streetBase.b;
            sc.a = conc * (surge ? 0.85f : 0.45f);
            streetImgs[i].color = sc;
            phase[i] += Time.deltaTime * (0.6f + conc * 1.2f);
            var pos = street[i].anchoredPosition;
            pos.x -= (26f + conc * 58f) * Time.deltaTime;
            pos.y = -100f + Mathf.Sin(phase[i]) * 26f;
```

```

        if (pos.x < -520f) pos.x = -350f;
        street[i].anchoredPosition = pos;
        float s = 26f + Mathf.Sin(phase[i] * 0.8f) * 6f + (surge ? 10f : 0f);
        street[i].sizeDelta = new Vector2(s, s);
    }
    yield return null;
}

static void ResetPuff(RectTransform puff, float delayX)
{
    puff.anchoredPosition = new Vector2(-350 + (delayX * 40f) % 160f, -40f);
}

// ----- HUD -----
public void SetCoins(int c) => _coinText.text = "    " + c;
public void SetHearts(int h) => _heartText.text = "    " + h;
public void SetHint(string hint) => _hintText.text = hint;
public void HideOpeningOverlay() => _openingOverlay.gameObject.SetActive(false);
public void ShowCraftingPanel(bool show) => _craftingPanel.gameObject.SetActive(show);
public void ShowBubble(bool show) => _bubble.gameObject.SetActive(show);
public void ShowVipBanner(bool show) => _vipBanner.gameObject.SetActive(show);
public void ShowSettle(int coins, int hearts, int stars, int codex)
{
    _settlePanel.gameObject.SetActive(true);
    _settleStats.text = $"    {coins} ·    {hearts} ·    {stars} ·    {codex} \n    {NewbieConfig.RecipeNam
}

// -----
RectTransform MakeCanvasChild(string name, Color color)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(_canvas.transform, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    var img = rt.GetComponent<Image>();
    img.sprite = _square;
    img.color = color;
    return rt;
}

RectTransform MakeChildOf(RectTransform parent, string name, Vector2 pos, Vector2 size)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = size;
    return rt;
}

RectTransform MakeOverlayPanel(string name, Color color)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));

```

```
        go.transform.SetParent(_canvas.transform, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = Vector2.zero;
        rt.anchorMax = Vector2.one;
        rt.offsetMin = rt.offsetMax = Vector2.zero;
        var image = go.GetComponent<Image>();
        image.sprite = _square;
        image.color = color;
        return rt;
    }
    /// <summary>          + </summary>
    RectTransform MakeOverlayBubble()
    {
        var rt = MakeCanvasChild("Bubble", MilkWhite);
        rt.anchoredPosition = new Vector2(0, 180);
        rt.sizeDelta = new Vector2(460, 220);
        return rt;
    }
    Button MakePanelButton(RectTransform parent, string label, Vector2 pos, Action onClick)
    {
        var go = new GameObject(label, typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(300, 96);
        var image = go.GetComponent<Image>();
        image.sprite = _square;
        image.color = SageGreen;
        MakeText(go.transform, label, 42, Vector2.zero, TextAnchor.MiddleCenter, 300, 96).color = MilkWhite;
        var button = go.GetComponent<Button>();
        button.onClick.AddListener(() => onClick());
        return button;
    }
    static Text MakeText(Transform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
    {
        var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(w, h);
        var t = go.GetComponent<Text>();
        t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        t.text = text;
        t.fontSize = size;
        t.alignment = anchor;
        t.color = new Color(0.91f, 0.89f, 0.85f);
        t.supportRichText = true;
        return t;
    }
```

```

    }
    static Sprite MakeSquareSprite()
    {
        var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
        var px = tex.GetPixels32();
        for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
        tex.SetPixels32(px);
        tex.Apply(false, false);
        return Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
    }
}

// SignDoubleOracle.cs
using System;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// T014-B slice 6: the sign_double consumption law (ad point table
    /// row "sign_double", V1.2 cap 1/day, face = today's daily sign-in
    /// still unclaimed). The grant doubles that day's sign-in reward
    /// deposit - the IAA upsell rides the D1-D7 retention moment. The
    /// oracle owns the shared AdCapPolicy consume exactly like the book
    /// doubler (P06's mock ad service has no cap logic); it stays pure C#
    /// with injected day keys so the whole law is headless-verifiable.
    /// The tutorial keeps the doubler face hidden (zero-ad red line
    /// before M5); the formal sign-in batch lights it via
    /// SetDoubleFace(true). Placement ruling: the sign-in sheet lives on
    /// the battle HUD because the main screen stays ad-free by design
    /// (MainUI owns no ad slots).
    /// </summary>
    public sealed class SignDoubleOracle
    {
        readonly AdPointEntry _point;
        readonly Sy.Platform.AdCapPolicy _adCap;
        public SignDoubleOracle(AdPointLaw law)
            : this(law, Sy.Platform.AdCapPolicy.DefaultDailyCap) { }
        /// <summary>Gates construct oracles with a small global cap to pin
        /// the global/category interplay without touching the policy type
        /// from the editor assembly (book-doubler pattern).</summary>
        public SignDoubleOracle(AdPointLaw law, int globalDailyCap)
        {
            _point = default;
            var pts = law.Points;
            for (int i = 0; i < pts.Length; i++)
                if (pts[i].Id == "sign_double") { _point = pts[i]; break; }
            _adCap = new Sy.Platform.AdCapPolicy(globalDailyCap);
        }
        public string PointId => _point.Id;
        public string Category => _point.Category;
        public int DailyCap => _point.CapPerDay;
    }
}

```

```

    public int UsedToday(string dayKey)
        => _adCap.UsedCategoryToday(dayKey, _point.Category);
    /// <summary>Pre-offer gate: only an unclaimed daily sign-in with
    /// category budget left - no doubler on a signed day, no
    /// dead-end taps.</summary>
    public bool CanOffer(string dayKey, bool signInPendingToday)
        => ValidFace(signInPendingToday)
        && _adCap.HasCategoryBudget(dayKey, _point.Category);
    /// <summary>Watched-ad grant: consumes one sign_double slot
    /// (1/day) and reports success - the caller then signs in at
    /// double value. Denies without consuming whenever the face is
    /// invalid (already signed) or the day's budget is gone.</summary>
    public bool TryGrant(string dayKey, bool signInPendingToday)
    {
        if (_point.Id == null) return false;
        if (!ValidFace(signInPendingToday)) return false;
        return _adCap.TryConsumeRewarded(dayKey, _point.Category);
    }
    static bool ValidFace(bool signInPendingToday) => signInPendingToday;
}
}
// SignInPanel.cs
using System;
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P06.Newbie
{
    /// <summary>Live face state of the daily sign-in: whether today is
    /// still unclaimed, the reward a sign-in would pay, the saved streak
    /// and the streak a sign-in would produce. Pure snapshot - the gate
    /// pins the builder law, the panel only renders it.</summary>
    public readonly struct SignInFaceModel
    {
        public readonly bool PendingToday;
        public readonly long TodayReward;
        public readonly int CurrentStreak;
        public readonly int NextStreak;
        public SignInFaceModel(bool pendingToday, long todayReward, int currentStreak, int nextStreak)
        {
            PendingToday = pendingToday;
            TodayReward = todayReward;
            CurrentStreak = currentStreak;
            NextStreak = nextStreak;
        }
    }
    /// <summary>
    /// T014-B slice 6: the daily sign-in sheet (the sign_double ad
    /// point's consumption surface). Placement ruling: the sheet opens
    /// from the battle HUD because the main screen stays ad-free by
    /// design (MainUI owns no ad slots) - the sign-in streak is the

```

```

/// D1-D7 retention loop and the doubler rides the claim moment.
/// Same split as the codex page: the panel is presentation-only,
/// BuildModel is the pure static law the gate asserts, every display
/// string resolves through the shipped strings table (no Chinese in
/// source) and the panel never grants by itself - the launcher owns
/// the wallet deposit, the save write and the ad funnel.
/// <summary>
public sealed class SignInPanel : MonoBehaviour
{
    public const string KeyTitle = "signin_panel_title";
    public const string KeyStreakLabel = "signin_streak_label";
    public const string KeyRewardLabel = "signin_reward_label";
    public const string KeyClaimButton = "signin_claim_button";
    public const string KeyClaimedLabel = "signin_claimed_label";
    public const string KeyCloseLabel = "signin_close_label";
    public const string KeyDoubleButton = "sign_double_button";
    public event Action SignInRequested;
    public event Action DoubleSignInRequested;
    RectTransform _overlay;
    RectTransform _sheetRoot;
    bool _doubleFace;
    static readonly Color SageGreen = new Color(0.561f, 0.659f, 0.525f); // grey-green #8FA886
    static readonly Color MilkWhite = new Color(0.961f, 0.941f, 0.910f); // milk white #F5F0E8
    static readonly Color OchreBrown = new Color(0.722f, 0.525f, 0.420f); // ochre #B8866B
    static readonly Color ColdGrey = new Color(0.620f, 0.660f, 0.720f); // cold Morandi grey-blue
    static readonly Color InkBrown = new Color(0.45f, 0.36f, 0.30f); // ink-brown copy
    /// <summary>Pure face law: the sign-in result of (lastDay, today,
    /// streak) over the reward table - the same SignInSystem law the
    /// launcher claims through, so the sheet can never disagree with
    /// the wallet. today == lastDay (&gt;= 0) means already signed.</summary>
    public static SignInFaceModel BuildModel(long lastSignInDayIndex, long todayIndex, int streak, long[] rewardTable)
    {
        var r = new Sy.Idle.SignInSystem(rewardTable).SignIn(lastSignInDayIndex, todayIndex, streak);
        return new SignInFaceModel(r.Accepted, r.Reward, streak, r.Streak);
    }
    public void Initialize(Canvas canvas)
    {
        _overlay = new GameObject("SignInOverlay", typeof(RectTransform), typeof(Image)).GetComponent<RectTransform>();
        _overlay.transform.SetParent(canvas.transform, false);
        _overlay.anchorMin = Vector2.zero;
        _overlay.anchorMax = Vector2.one;
        _overlay.offsetMin = _overlay.offsetMax = Vector2.zero;
        var dim = _overlay.GetComponent<Image>();
        dim.sprite = MakeSquareSpriteShared();
        dim.color = new Color(0.35f, 0.30f, 0.27f, 0.92f);
        var paper = MakeCard(_overlay, "SignInPaper", Vector2.zero, new Vector2(940, 900),
            new Color(MilkWhite.r, MilkWhite.g, MilkWhite.b, 0.97f));
        _sheetRoot = MakeContainer(paper, "SheetRoot");
        _overlay.gameObject.SetActive(false);
    }
}

```

```
/// &summaryOpens the sheet with a fresh face snapshot pushed by
/// the launcher (the launcher owns the save + clock reads).&/summary
public void Open(SignInFaceModel model)
{
    if (_overlay == null) return;
    _overlay.gameObject.SetActive(true);
    Refresh(model);
}
/// &summaryArms or hides the sign_double doubler face (T014-B:
/// the launcher locks it false through the tutorial - zero-ad red
/// line before M5 - and the formal sign-in batch lights it).&/summary
public void SetDoubleFace(bool armed)
{
    _doubleFace = armed;
}
/// &summaryRe-renders the sheet from a face snapshot so the
/// claim state, streak and doubler never go stale.&/summary
public void Refresh(SignInFaceModel model)
{
    if (_overlay == null) return;
    Clear(_sheetRoot);
    MakeText(_sheetRoot, StoryBoard.Text(KeyTitle), 72, new Vector2(0, 300),
        TextAnchor.MiddleCenter, 940, 110).color = OchreBrown;
    MakeText(_sheetRoot, string.Format(StoryBoard.Text(KeyStreakLabel), model.CurrentStreak),
        40, new Vector2(0, 190), TextAnchor.MiddleCenter, 900, 70).color = InkBrown;
    // reward line: an unclaimed day shows what signing in pays,
    // a signed day shows the come-back-tomorrow copy
    MakeText(_sheetRoot, model.PendingToday
        ? string.Format(StoryBoard.Text(KeyRewardLabel), model.TodayReward)
        : StoryBoard.Text(KeyClaimedLabel),
        46, new Vector2(0, 90), TextAnchor.MiddleCenter, 900, 84).color = InkBrown;
    var claim = MakeButton(_sheetRoot,
        model.PendingToday ? StoryBoard.Text(KeyClaimButton) : StoryBoard.Text(KeyClaimedLabel),
        new Vector2(0, -40), new Vector2(640, 110), model.PendingToday);
    if (model.PendingToday)
        claim.onClick.AddListener(() => SignInRequested?.Invoke());
    // T014-B sign_double doubler: visible only when the formal
    // face is armed and today is actually unclaimed - the
    // tutorial never shows an ad button (zero-ad red line before
    // M5) and a signed day shows no dead upsell.
    bool showDouble = _doubleFace && model.PendingToday;
    if (showDouble)
    {
        MakeButton(_sheetRoot, StoryBoard.Text(KeyDoubleButton),
            new Vector2(0, -160), new Vector2(640, 72), true)
            .onClick.AddListener(() => DoubleSignInRequested?.Invoke());
    }
    MakeButton(_sheetRoot, StoryBoard.Text(KeyCloseLabel),
        new Vector2(0, showDouble ? -268 : -200), new Vector2(420, 92), true)
        .onClick.AddListener(() => _overlay.gameObject.SetActive(false));
}
```

```
}
// —— helpers (programmatic UGUI, zero scene-YAML) ——
static void Clear(RectTransform root)
{
    for (int i = root.childCount - 1; i >= 0; i--)
        Destroy(root.GetChild(i).gameObject);
}
static RectTransform MakeContainer(RectTransform parent, string name)
{
    var go = new GameObject(name, typeof(RectTransform));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = Vector2.zero;
    rt.offsetMin = rt.offsetMax = Vector2.zero;
    return rt;
}
static RectTransform MakeCard(RectTransform parent, string name, Vector2 pos, Vector2 size, Color color)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = size;
    var img = go.GetComponent<Image>();
    img.sprite = MakeSquareSpriteShared();
    img.color = color;
    return rt;
}
static Text MakeText(RectTransform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
{
    var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(w, h);
    var t = go.GetComponent<Text>();
    t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
    t.text = text;
    t.fontSize = size;
    t.alignment = anchor;
    t.color = InkBrown;
    return t;
}
static Button MakeButton(RectTransform parent, string label, Vector2 pos, Vector2 size, bool enabled)
{
    var go = new GameObject("ActionButton", typeof(RectTransform), typeof(Image), typeof(Button));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
```

```

        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        var img = go.GetComponent<Image>();
        img.sprite = MakeSquareSpriteShared();
        img.color = enabled ? SageGreen : new Color(ColdGrey.r, ColdGrey.g, ColdGrey.b, 0.55f);
        var txt = MakeText(rt, label, 38, Vector2.zero, TextAnchor.MiddleCenter, size.x, size.y);
        txt.color = MilkWhite;
        txt.raycastTarget = false;
        return go.GetComponent<Button>();
    }
    static Sprite _sharedSquare;
    static Sprite MakeSquareSpriteShared()
    {
        if (_sharedSquare == null)
        {
            var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
            var px = tex.GetPixels32();
            for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
            tex.SetPixels32(px);
            tex.Apply(false, false);
            _sharedSquare = Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
        }
        return _sharedSquare;
    }
}

/// <summary>Daily sign-in persistence (red line 12: local
/// PlayerPrefs authority, zero server). Format: "lastDay,streak",
/// empty = never signed.</summary>
public static class SignInSave
{
    public const string StateKey = "p06_signin_state";
    public static void Load(out long lastSignInDayIndex, out int streak)
    {
        lastSignInDayIndex = -1;
        streak = 0;
        var raw = UnityEngine.PlayerPrefs.GetString(StateKey, "");
        if (string.IsNullOrEmpty(raw)) return;
        var parts = raw.Split(',');
        if (parts.Length >= 1 && long.TryParse(parts[0], out var d)) lastSignInDayIndex = d;
        if (parts.Length >= 2 && int.TryParse(parts[1], out var s)) streak = s;
    }
    public static void Save(long lastSignInDayIndex, int streak)
    {
        UnityEngine.PlayerPrefs.SetString(StateKey, lastSignInDayIndex + "," + streak);
        UnityEngine.PlayerPrefs.Save();
    }
}

/// <summary>Test probe: wipes the persisted sign-in state.</summary>
public static void ClearForTests()
{

```

```
        UnityEngine.PlayerPrefs.DeleteKey(StateKey);
    }
}
}
// StoryBoard.cs
using System;
using System.Collections.Generic;
using UnityEngine;
namespace Sy.P06.Newbie
{
    /// <summary>One row of the story list: a chapter plus its live/locked
    /// state under the current week (weekly flip driver).</summary>
    public readonly struct ChapterRow
    {
        public readonly ChapterEntry Chapter;
        public readonly bool IsLive;
        public ChapterRow(ChapterEntry chapter, bool isLive)
        {
            Chapter = chapter;
            IsLive = isLive;
        }
    }
    /// <summary>
    /// T009 story service + weekly flip driver. Joins GuestConfig chapters
    /// with RecipeConfig clue ids and the shipped strings table, and
    /// resolves which chapters are live for the current week. The week
    /// source is an injectable provider (live builds can hook the calendar
    /// week; gates pin exact weeks) so the whole release curve stays
    /// headless-verifiable. Display text comes from the cfg_i18n TextAsset
    /// shipped under Resources (authoring table: Design/configs/P06/
    /// cfg_i18n.json; GuestCheck U1 pins the two byte-for-byte) so runtime
    /// code never hardcodes display copy.
    /// </summary>
    public static class StoryBoard
    {
        static Func<int> _weekProvider = DefaultWeek;
        static Dictionary<string, string> _strings;
        static bool _stringsLoaded;
        static int DefaultWeek() => 0;
        /// <summary>Pins the week source (gates/tests). Passing null
        /// restores the default provider.</summary>
        public static void SetWeekProvider(Func<int> provider)
        {
            _weekProvider = provider ?? DefaultWeek;
        }
        public static int CurrentWeek => _weekProvider();
        /// <summary>Weekly flip driver: a chapter is live when playable at
        /// the current week (stock chapters flip in place at their
        /// releaseWeek and never regress).</summary>
        public static bool IsLive(ChapterEntry chapter)
```

```
{
    return GuestConfig.IsPlayableAt(chapter, CurrentWeek);
}
public static int LiveChapterCount()
{
    return GuestConfig.CountPlayableAt(GuestConfig.ChaptersBuiltIn(), CurrentWeek);
}
/// <summary>Story list for one guest: every chapter they host with
/// its live/locked state (locked rows keep releaseWeek so the UI
/// can show the "next week" label).</summary>
public static List<ChapterRow> RowsForGuest(int guestId)
{
    var rows = new List<ChapterRow>();
    var chapters = GuestConfig.ChaptersBuiltIn();
    for (int i = 0; i < chapters.Length; i++)
        if (chapters[i].GuestId == guestId)
            rows.Add(new ChapterRow(chapters[i], IsLive(chapters[i])));
    return rows;
}
/// <summary>Resolved clue lines for a chapter, in clue order:
/// finished display sentences from the strings table, ready for the
/// story UI (no raw keys, no manual joins).</summary>
public static List<string> ClueLines(ChapterEntry chapter)
{
    var lines = new List<string>();
    for (int i = 0; i < chapter.ClueIds.Length; i++)
        lines.Add(Text("clue_" + chapter.ClueIds[i] + "_text"));
    return lines;
}
/// <summary>Display text lookup; falls back to the raw key so a
/// missing string stays visible and debuggable instead of blank.</summary>
public static string Text(string key)
{
    return TryText(key, out var value) ? value : key;
}
public static bool TryText(string key, out string value)
{
    if (!_stringsLoaded) LoadStrings();
    if (_strings != null && _strings.TryGetValue(key, out value)) return true;
    value = null;
    return false;
}
static void LoadStrings()
{
    _stringsLoaded = true;
    var asset = Resources.Load<TextAsset>("cfg_i18n");
    if (asset == null) return;
    GuestConfig.TryI18nKeysFromJson(asset.text, out _strings);
}
}
```

```
}
// StoryCardComposer.cs
using Sy.Social;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// T015-D story-completion card: the chapter the player just finished
    /// becomes a 3:4 keepsake png (K engine reuse, xiaohongshu-ready share
    /// of the three-piece set). The face carries the chapter title, the
    /// host guest and the closing line - real story data only (the season
    /// panorama slice waits for its art + season mechanics; no fabricated
    /// seasons on a shipped face). Copy resolves through the i18n table at
    /// compose time; the query carries the chapter id for share attribution.
    /// </summary>
    public static class StoryCardComposer
    {
        public const string ShareQueryPrefix = "src=share&story=";
        public const string TaglineKey = "story_card_line";
        public static ShareCard FromChapter(ChapterEntry chapter)
        {
            string title = StoryBoard.Text(chapter.TitleKey);
            string guest = StoryBoard.Text(GuestNameKeyOf(chapter.GuestId));
            string close = StoryBoard.Text(chapter.CloseKey);
            return AutoCardComposer.Card(
                ScentCardComposer.RecipeTemplateKey, // shared wood-paper brand face
                title,
                new[] { guest, close, StoryBoard.Text(TaglineKey) },
                StoryBoard.Text(ScentCardComposer.FooterKey),
                ShareQueryPrefix + chapter.Id);
        }
        /// <summary>Id resolution with the no-dead-end law: an unknown id
        /// falls back to the first live chapter so the share face never
        /// renders an empty promise.</summary>
        public static ShareCard FromChapterId(int chapterId)
        {
            var chapters = GuestConfig.ChaptersBuiltIn();
            for (int i = 0; i < chapters.Length; i++)
                if (chapters[i].Id == chapterId) return FromChapter(chapters[i]);
            return chapters.Length > 0 ? FromChapter(chapters[0]) : null;
        }
        static string GuestNameKeyOf(int guestId)
        {
            var guests = GuestConfig.GuestsBuiltIn();
            for (int i = 0; i < guests.Length; i++)
                if (guests[i].Id == guestId) return guests[i].NameKey;
            return string.Empty;
        }
    }
}
// StoryPanel.cs
```

```
using System;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P06.Newbie
{
    /// <summary>One story-list section: a guest host plus their chapter
    /// rows under the current week (live or locked).</summary>
    public readonly struct GuestSection
    {
        public readonly GuestEntry Guest;
        public readonly List<ChapterRow> Rows;
        public GuestSection(GuestEntry guest, List<ChapterRow> rows)
        {
            Guest = guest;
            Rows = rows;
        }
    }
    /// <summary>
    /// T009 story panel - the in-game wiring for the StoryBoard service
    /// (RowsForGuest / ClueLines / Text). Programmatic UGUI in the Morandi
    /// palette, zero scene-YAML drift, presentation-only (never touches the
    /// shop state machine). Every display string resolves through the
    /// shipped strings table so no Chinese lives in source; the list layer
    /// is a pure static model (BuildSections) the headless gate asserts
    /// without rendering. Locked rows keep their releaseWeek so the UI can
    /// promise when the chapter flips live.
    /// </summary>
    public sealed class StoryPanel : MonoBehaviour
    {
        public const string KeyTitle = "story_panel_title";
        public const string KeyWeeklyHint = "story_weekly_hint";
        public const string KeyLockedLabel = "story_locked_label";
        public const string KeyCloseLabel = "story_close_label";
        public const string KeyPreviewButton = "story_preview_button";
        public const string KeyPreviewNote = "story_preview_note";
        public const string KeyStoryShareButton = "story_share_button";
        /// <summary>T014-B3: a locked row's preview button fired. The launcher
        /// owns the ad funnel (offer/ad_fail/accept/blocked); the panel only
        /// renders - it never grants.</summary>
        public event Action<ChapterEntry> PreviewTapped;
        /// <summary>T015-D: the chapter-detail share button fired. The
        /// launcher owns the compose + platform share funnel; the panel only
        /// renders the face it was armed with.</summary>
        public event Action<ChapterEntry> StoryShareTapped;
        bool _previewArmed;    // T014-B3: locked rows show the preview button only when armed
        bool _storyShareArmed; // T015-D: live chapter details show the share button only when armed
        Sprite _square;
        RectTransform _overlay;
        RectTransform _listRoot;
```

```

RectTransform _detailRoot;
static readonly Color SageGreen = new Color(0.561f, 0.659f, 0.525f); // #8FA886
static readonly Color MilkWhite = new Color(0.961f, 0.941f, 0.910f); // #F5F0E8
static readonly Color OchreBrown = new Color(0.722f, 0.525f, 0.420f); // #B8866B
static readonly Color ColdGrey = new Color(0.620f, 0.660f, 0.720f); // .
static readonly Color InkBrown = new Color(0.45f, 0.36f, 0.30f); //
/// <summary>Pure list model for the gate and the view: one section
/// per guest that hosts chapters, each with their rows under the
/// current week. Guests without chapters stay out of the list
/// (no dead UI, no fake promises).</summary>
public static List<GuestSection> BuildSections()
{
    var sections = new List<GuestSection>();
    var guests = GuestConfig.GuestsBuiltIn();
    for (int i = 0; i < guests.Length; i++)
    {
        var rows = StoryBoard.RowsForGuest(guests[i].Id);
        if (rows.Count > 0)
            sections.Add(new GuestSection(guests[i], rows));
    }
    return sections;
}
public void Initialize(Canvas canvas)
{
    _square = MakeSquareSpriteShared();
    _overlay = new GameObject("StoryOverlay", typeof(RectTransform), typeof(Image)).GetComponent<RectTransform>();
    _overlay.transform.SetParent(canvas.transform, false);
    _overlay.anchorMin = Vector2.zero;
    _overlay.anchorMax = Vector2.one;
    _overlay.offsetMin = _overlay.offsetMax = Vector2.zero;
    var dim = _overlay.GetComponent<Image>();
    dim.sprite = _square;
    dim.color = new Color(0.35f, 0.30f, 0.27f, 0.92f);
    var paper = MakeCard(_overlay, "StoryPaper", Vector2.zero, new Vector2(1020, 1700),
        new Color(MilkWhite.r, MilkWhite.g, MilkWhite.b, 0.97f));
    _listRoot = MakeContainer(paper, "ListRoot");
    _detailRoot = MakeContainer(paper, "DetailRoot");
    _detailRoot.gameObject.SetActive(false);
    _overlay.gameObject.SetActive(false);
}
/// <summary>Opens the panel on the chapter list, rebuilt from the
/// current week so live/locked states are always fresh.</summary>
public void Open()
{
    _overlay.gameObject.SetActive(true);
    ShowList();
}
/// <summary>T014-B3: arms/disarms the locked-row preview face. The
/// launcher keeps it locked in the tutorial (zero-ad red line before
/// M5); the formal story batch lights it.</summary>

```

```
public void SetPreviewFace(bool armed)
{
    _previewArmed = armed;
}
/// <summary>T015-D: arms/disarms the chapter-detail share face. The
/// launcher arms it only when a platform service can carry the
/// payload (no dead buttons); sharing is not an ad point, so the
/// tutorial face may light too.</summary>
public void SetStoryShareFace(bool armed)
{
    _storyShareArmed = armed;
}
void ShowList()
{
    _detailRoot.gameObject.SetActive(false);
    Clear(_listRoot);
    _listRoot.gameObject.SetActive(true);
    MakeText(_listRoot, StoryBoard.Text(KeyTitle), 76, new Vector2(0, 690),
        TextAnchor.MiddleCenter, 1020, 110).color = OchreBrown;
    MakeText(_listRoot, StoryBoard.Text(KeyWeeklyHint), 32, new Vector2(0, 588),
        TextAnchor.MiddleCenter, 1020, 60).color = new Color(0.62f, 0.55f, 0.50f);
    float y = 440f;
    var sections = BuildSections();
    for (int s = 0; s < sections.Count; s++)
    {
        var section = sections[s];
        MakeText(_listRoot, StoryBoard.Text(section.Guest.NameKey), 42, new Vector2(0, y),
            TextAnchor.MiddleLeft, 940, 64).color = OchreBrown;
        MakeText(_listRoot, StoryBoard.Text(section.Guest.PersonaKey), 28, new Vector2(0, y - 54),
            TextAnchor.MiddleLeft, 940, 46).color = new Color(0.62f, 0.55f, 0.50f);
        y -= 130f;
        for (int r = 0; r < section.Rows.Count; r++)
        {
            var row = section.Rows[r];
            var chapter = row.Chapter;
            var nameKey = section.Guest.NameKey;
            var card = MakeCard(_listRoot, "Chapter_" + chapter.Id, new Vector2(0, y),
                new Vector2(940, 96), row.IsLive
                ? new Color(SageGreen.r, SageGreen.g, SageGreen.b, 0.16f)
                : new Color(ColdGrey.r, ColdGrey.g, ColdGrey.b, 0.22f));
            if (row.IsLive)
            {
                var accent = MakeCard(card, "Accent", new Vector2(-452, 0), new Vector2(8, 72), SageGreen);
                var img = accent.GetComponent<Image>();
                img.raycastTarget = false;
                MakeText(card, StoryBoard.Text(chapter.TitleKey), 38, new Vector2(-60, 0),
                    TextAnchor.MiddleLeft, 640, 88).color = InkBrown;
                var button = card.gameObject.AddComponent<Button>();
                button.onClick.AddListener(() => ShowChapter(nameKey, chapter));
            }
        }
    }
}
```

```

        else
        {
            MakeText(card, StoryBoard.Text(chapter.TitleKey), 38, new Vector2(-60, 0),
                TextAnchor.MiddleLeft, 640, 88).color = new Color(0.56f, 0.55f, 0.53f);
            if (_previewArmed)
            {
                // T014-B3: armed face - the release-week label
                // shifts left, the preview button takes the right
                // rail; only locked rows carry it (live rows open
                // free, dead ends never render).
                MakeText(card, string.Format(StoryBoard.Text(KeyLockedLabel), chapter.ReleaseWeek),
                    26, new Vector2(-160, 22), TextAnchor.MiddleLeft, 440, 44).color = ColdGrey;
                var tapped = chapter;
                MakeButton(card, StoryBoard.Text(KeyPreviewButton), new Vector2(330, 0),
                    new Vector2(210, 72), () => PreviewTapped?.Invoke(tapped));
            }
            else
            {
                MakeText(card, string.Format(StoryBoard.Text(KeyLockedLabel), chapter.ReleaseWeek),
                    30, new Vector2(270, 0), TextAnchor.MiddleRight, 360, 60).color = ColdGrey;
            }
        }
        y -= 110f;
    }
    y -= 10f;
}

MakeButton(_listRoot, StoryBoard.Text(KeyCloseLabel), new Vector2(0, -770),
    new Vector2(420, 92), () => _overlay.gameObject.SetActive(false));
}

void ShowChapter(string guestNameKey, ChapterEntry chapter)
{
    _listRoot.gameObject.SetActive(false);
    Clear(_detailRoot);
    _detailRoot.gameObject.SetActive(true);
    MakeText(_detailRoot, StoryBoard.Text(chapter.TitleKey), 62, new Vector2(0, 700),
        TextAnchor.MiddleCenter, 1020, 120).color = OchreBrown;
    MakeText(_detailRoot, "— " + StoryBoard.Text(guestNameKey), 34, new Vector2(0, 560),
        TextAnchor.MiddleCenter, 1020, 60).color = new Color(0.55f, 0.48f, 0.42f);
    MakeText(_detailRoot, StoryBoard.Text(chapter.OpenKey), 40, new Vector2(0, 420),
        TextAnchor.MiddleCenter, 940, 220).color = InkBrown;
    float y = 150f;
    var lines = StoryBoard.ClueLines(chapter);
    for (int i = 0; i < lines.Count; i++)
    {
        MakeText(_detailRoot, lines[i], 32, new Vector2(0, y),
            TextAnchor.MiddleCenter, 940, 72).color = InkBrown;
        y -= 80f;
    }
    var closeLine = MakeText(_detailRoot, StoryBoard.Text(chapter.CloseKey), 40, new Vector2(0, y - 110),
        TextAnchor.MiddleCenter, 940, 150);
}

```

```

        closeLine.color = new Color(0.72f, 0.58f, 0.55f);
        // T015-D: live-chapter keepsake share - the armed face splits the
        // bottom rail with the close button; disarmed keeps the centered
        // close layout (the face never renders a dead button).
        if (_storyShareArmed)
        {
            var tapped = chapter;
            MakeButton(_detailRoot, StoryBoard.Text(KeyCloseLabel), new Vector2(-250, -790),
                new Vector2(420, 84), ShowList);
            MakeButton(_detailRoot, StoryBoard.Text(KeyStoryShareButton), new Vector2(250, -790),
                new Vector2(420, 84), () => StoryShareTapped?.Invoke(tapped));
        }
        else
        {
            MakeButton(_detailRoot, StoryBoard.Text(KeyCloseLabel), new Vector2(0, -790),
                new Vector2(420, 84), ShowList);
        }
    }

    /// <summary>T014-B3: renders an ad-granted locked-chapter preview -
    /// title, host guest and the opening lines only. Clues and the close
    /// line stay sealed until the chapter flips live (the preview never
    /// spoils the weekly reveal), and the note names the flavor so the
    /// player always knows this is a peek, not the full chapter.</summary>
    public void ShowPreview(ChapterEntry chapter)
    {
        _listRoot.gameObject.SetActive(false);
        Clear(_detailRoot);
        _detailRoot.gameObject.SetActive(true);
        MakeText(_detailRoot, StoryBoard.Text(chapter.TitleKey), 62, new Vector2(0, 700),
            TextAnchor.MiddleCenter, 1020, 120).color = OchreBrown;
        MakeText(_detailRoot, StoryBoard.Text(KeyPreviewNote), 30, new Vector2(0, 566),
            TextAnchor.MiddleCenter, 1020, 56).color = ColdGrey;
        MakeText(_detailRoot, "— " + StoryBoard.Text(GuestNameKeyOf(chapter.GuestId)), 34,
            new Vector2(0, 486), TextAnchor.MiddleCenter, 1020, 60).color = new Color(0.55f, 0.48f, 0.42f);
        MakeText(_detailRoot, StoryBoard.Text(chapter.OpenKey), 40, new Vector2(0, 300),
            TextAnchor.MiddleCenter, 940, 240).color = InkBrown;
        MakeButton(_detailRoot, StoryBoard.Text(KeyCloseLabel), new Vector2(0, -790),
            new Vector2(420, 84), ShowList);
    }

    static string GuestNameKeyOf(int guestId)
    {
        var guests = GuestConfig.GuestsBuiltIn();
        for (int i = 0; i < guests.Length; i++)
            if (guests[i].Id == guestId) return guests[i].NameKey;
        return string.Empty;
    }

    // ——— UGUI· YAML ———
    static void Clear(RectTransform root)
    {
        for (int i = root.childCount - 1; i >= 0; i--)

```

```
        Destroy(root.GetChild(i).gameObject);
    }
    static RectTransform MakeContainer(RectTransform parent, string name)
    {
        var go = new GameObject(name, typeof(RectTransform));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = Vector2.zero;
        rt.offsetMin = rt.offsetMax = Vector2.zero;
        return rt;
    }
    static RectTransform MakeCard(RectTransform parent, string name, Vector2 pos, Vector2 size, Color color)
    {
        var go = new GameObject(name, typeof(RectTransform), typeof(Image));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        var img = go.GetComponent<Image>();
        img.sprite = MakeSquareSpriteShared();
        img.color = color;
        return rt;
    }
    static Text MakeText(RectTransform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
    {
        var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = new Vector2(w, h);
        var t = go.GetComponent<Text>();
        t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        t.text = text;
        t.fontSize = size;
        t.alignment = anchor;
        t.color = InkBrown;
        return t;
    }
    static void MakeButton(RectTransform parent, string label, Vector2 pos, Vector2 size, Action onClick)
    {
        var go = new GameObject("ActionButton", typeof(RectTransform), typeof(Image), typeof(Button));
        go.transform.SetParent(parent, false);
        var rt = (RectTransform)go.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
        rt.anchoredPosition = pos;
        rt.sizeDelta = size;
        var img = go.GetComponent<Image>();
        img.sprite = MakeSquareSpriteShared();
```

```

        img.color = SageGreen;
        var txt = MakeText(rt, label, 40, Vector2.zero, TextAnchor.MiddleCenter, size.x, size.y);
        txt.color = MilkWhite;
        txt.raycastTarget = false;
        go.GetComponent<Button>().onClick.AddListener(() => onClick());
    }
    static Sprite _sharedSquare;
    static Sprite MakeSquareSpriteShared()
    {
        if (_sharedSquare == null)
        {
            var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
            var px = tex.GetPixels32();
            for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
            tex.SetPixels32(px);
            tex.Apply(false, false);
            _sharedSquare = Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
        }
        return _sharedSquare;
    }
}

// StoryPreviewOracle.cs
using System;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// T014-B slice 3: the story_preview consumption law (ad point table row
    /// "story_preview", V1.2 cap 1/day, surface story/locked chapter). The
    /// grant flips a locked chapter into a read-only preview of its opening
    /// - the content-style rewarded point the pitch doc calls out for the
    /// female-leaning audience (higher ad tolerance on story content). The
    /// oracle owns the shared AdCapPolicy consume exactly like the speed
    /// oracle (P06's mock ad service has no cap logic); it stays pure C#
    /// with injected day keys so the whole law is headless-verifiable. The
    /// tutorial keeps the preview face hidden (zero-ad red line before M5);
    /// the formal story batch lights it via SetPreviewFace(true).
    /// </summary>
    public sealed class StoryPreviewOracle
    {
        readonly AdPointEntry _point;
        readonly Sy.Platform.AdCapPolicy _adCap;
        public StoryPreviewOracle(AdPointLaw law)
            : this(law, Sy.Platform.AdCapPolicy.DefaultDailyCap) { }
        /// <summary>Gates construct oracles with a small global cap to pin
        /// the global/category interplay without touching the policy type
        /// from the editor assembly (CraftSpeedOracle pattern).</summary>
        public StoryPreviewOracle(AdPointLaw law, int globalDailyCap)
        {
            _point = default;

```

```

        var pts = law.Points;
        for (int i = 0; i < pts.Length; i++)
            if (pts[i].Id == "story_preview") { _point = pts[i]; break; }
        _adCap = new Sy.Platform.AdCapPolicy(globalDailyCap);
    }
    public string PointId => _point.Id;
    public string Category => _point.Category;
    public int DailyCap => _point.CapPerDay;
    public int UsedToday(string dayKey)
        => _adCap.UsedCategoryToday(dayKey, _point.Category);
    /// <summary>Pre-offer gate: only a locked chapter face with category
    /// budget left - no preview button on live rows, no dead-end taps.</summary>
    public bool CanOffer(string dayKey, bool chapterLocked)
        => ValidFace(chapterLocked)
        && _adCap.HasCategoryBudget(dayKey, _point.Category);
    /// <summary>Watched-ad grant: consumes the one story_preview slot
    /// (1/day) and reports success - the caller then opens the preview.
    /// Denies without consuming whenever the face is invalid (live
    /// chapter) or the day's budget is gone.</summary>
    public bool TryGrant(string dayKey, bool chapterLocked)
    {
        if (_point.Id == null) return false;
        if (!ValidFace(chapterLocked)) return false;
        return _adCap.TryConsumeRewarded(dayKey, _point.Category);
    }
    static bool ValidFace(bool chapterLocked) => chapterLocked;
}

// TipCounter.cs
using System;
using Sy.Core;
using Sy.Idle;
namespace Sy.P06.Newbie
{
    /// <summary>
    /// W11b offline tip counter: the shop keeps collecting tips while the
    /// player is away. Deterministic core - the clock arrives as a timestamp
    /// argument and the anchor lives in the local save slot (zero-server
    /// ruling: local save is authoritative), so the settle gate can replay
    /// any timeline. Fresh installs start the clock at first open and show
    /// zero fake tips (care/return principle: no fabricated earnings).
    /// Claiming slides the anchor forward (parallel claim, never a reset).
    /// </summary>
    public sealed class TipCounter
    {
        public const string AnchorSlot = "p06_idle_tip_anchor";
        readonly ISaveService _save;
        readonly FastClaimService _claim;
        public TipCounter(IdlerFormula formula, ISaveService save, long nowMs)
        {

```

```

        _save = save ?? throw new ArgumentNullException(nameof(save));
        long anchor = ParseAnchor(save.Load(AnchorSlot));
        if (anchor &lt;= 0)
        {
            anchor = nowMs;    // fresh install: the idle clock starts now
            _save.Save(AnchorSlot, anchor.ToString());
        }
        _claim = new FastClaimService(formula, anchor);
    }
    /// <summary>Unclaimed tips so far, without consuming them.</summary>
    public double PeekPending(long nowMs) => _claim.Peek(nowMs);
    /// <summary>Claim tips; the anchor slides to now (the clock never resets).</summary>
    public double Claim(long nowMs)
    {
        double amount = _claim.Claim(nowMs);
        _save.Save(AnchorSlot, _claim.AnchorTsMs.ToString());
        return amount;
    }
    static long ParseAnchor(string raw)
    {
        if (string.IsNullOrEmpty(raw)) return 0;
        return long.TryParse(raw, out long v) ? v : 0;
    }
}

// TutorialFlow.cs
using System;
namespace Sy.P06.Newbie
{
    /// <summary>
    ///     evt_newbie_step 1-5 ① ② ③ ④ ⑤ +
    ///     ShopSim      /Mastermind      / / —
    /// Play   headless   NewbieFlowCheck
    /// </summary>
    public sealed class TutorialFlow
    {
        public int Step { get; private set; }
        readonly Action<string> _hint;
        readonly Action<int> _onStep;
        public TutorialFlow(ShopSim sim, Action<string> hint, Action<int> onStep)
        {
            _hint = hint ?? delegate { };
            _onStep = onStep ?? delegate { };
            sim.OrderAccepted += OnOrderAccepted;
            sim.FeedbackGiven += OnFeedbackGiven;
            sim.BlendCompleted += OnBlendCompleted;
            sim.RecipeCardShown += OnRecipeCardShown;
            sim.OrderCompleted += OnOrderCompleted;
            sim.StarGained += OnStarGained;
            sim.CodexUnlocked += OnCodexUnlocked;

```

```

        sim.Settled += OnSettled;
        sim.StateChanged += OnStateChanged;
        _hint("      —      ");
    }
    void Advance(int step)
    {
        if (Step >= step) return;
        Step = step;
        _onStep(step);
    }
    void OnOrderAccepted()
    {
        Advance(1);
        _hint(NewbieConfig.GuestStoryClue); // §4.2 ②
    }
    void OnFeedbackGiven(int slot, int materialId, FeedbackState state, int trueSlot)
    {
        switch (state)
        {
            case FeedbackState.Correct:
                _hint(NewbieConfig.SlotNames[slot] + "      " + NewbieConfig.MaterialNames[materialId] + " —");
                break;
            case FeedbackState.Warm:
                _hint(string.Format(NewbieConfig.SnarkWarm, NewbieConfig.SlotNames[trueSlot])
                    + " " + NewbieConfig.MaterialNames[materialId] + " ");
                break;
            case FeedbackState.Cold:
                _hint(NewbieConfig.SnarkCold + " " + NewbieConfig.MaterialNames[materialId] + " ");
                break;
        }
    }
    void OnBlendCompleted()
    {
        Advance(2);
        _hint("      " + NewbieConfig.RecipeName + " — —");
    }
    void OnRecipeCardShown()
    {
        _hint("      — " + NewbieConfig.RecipeDesc + "      ");
    }
    void OnOrderCompleted(int coins, int hearts)
    {
        Advance(3);
        _hint("      +" + coins + " ·      +" + hearts + " —      ");
    }
    void OnStarGained(int stars)
    {
        _hint("      +" + NewbieConfig.OrderStars + "      " + stars + " —" + NewbieConfig.GuestName + "      ");
    }
    void OnCodexUnlocked()

```

```

        {
            Advance(4);
            _hint("          " + NewbieConfig.GuestName + "—          ");
        }
        void OnSettled()
        {
            Advance(5);
            _hint("          —          ");
        }
        void OnStateChanged(ShopSim.ShopState state)
        {
            switch (state)
            {
                case ShopSim.ShopState.GuestArriving:
                    _hint("          —          ");
                    break;
                case ShopSim.ShopState.WaitingAccept:
                    _hint(NewbieConfig.GuestName + "          " + NewbieConfig.RecipeName + " —          ");
                    break;
            }
        }
    }
}

// UiInputChainTests.cs
using System.Collections;
using NUnit.Framework;
using Sy.P06.Newbie;
using UnityEngine;
using UnityEngine.EventSystems;
using UnityEngine.InputSystem;
using UnityEngine.InputSystem.UI;
using UnityEngine.UI;
using UnityEngine.TestTools;
namespace Sy.P06.PmTests
{
    /// <summary>
    /// X1587 (U262 batch A, 12th doc sec 12 v2.0 input-chain truth law): the
    /// P06 direct-call debt absorption slice. VirtualUserCheck's persona bot
    /// fires the accept button's onClick.Invoke() straight on an inactive
    /// GameObject - the handler path only, skipping raycast, the active /
    /// interactable gate and the EventSystem. Preparing this injection domain
    /// flushed out the real bug that seam could never see: ShowBubble(true)
    /// had ZERO call sites in the whole project, so on a real device the
    /// tutorial told the player to tap a button that never appeared and the
    /// shop soft-locked at WaitingAccept (fixed in BattleLauncher.OnStateChanged
    /// the same batch). This PlayMode domain drives the SAME accept flow with
    /// SYNTHESIZED touch input through the real UGUI pointer chain, so the
    /// shipping seat - not the delegate - is what gets proven. Rig mirrors the
    /// editor BuildRig (WorldSpace 1080x1920 canvas + ortho camera + ShopSim
    /// + ShopView) plus the P08 X1297 rig additions.

```

```
/// X1597 (slice c, X1595 design freeze section 2): the N3 random-roam
/// bot law extends this domain - 72 seeded pokes across both blind
/// domains (in-face placeholder bands + off-face screen dead zones)
/// interleaved with discovered-seat roams (accept / drawer / HUD
/// affordances), every one through the real pointer chain; zero
/// softlock, first accept reachable via the true chain, the invest
/// book moving under the roaming finger, and the X1587 seat-swallow
/// law holding for the whole roam.
/// &lt;/summary&gt;
[TestFixture]
public sealed class UiInputChainTests : InputTestFixture
{
    GameObject _root;
    Canvas _canvas;
    Camera _cam;
    ShopSim _sim;
    ShopView _view;
    Button _acceptBtn;
    int _acceptEvents;
    int _acceptDone;
    Button[] _matButtons;
    Button _storyBtn, _caravanBtn, _codexBtn, _facilityBtn, _signBtn;
    int _materialEvents;
    int _investDone;
    const int Uilayer = 5;
    public override void Setup()
    {
        base.Setup();
        _root = new GameObject("UiInputChainRig");
        var esGo = new GameObject("TestEventSystem");
        esGo.transform.SetParent(_root.transform, false);
        esGo.AddComponent<EventSystem>();
        esGo.AddComponent<InputSystemUIInputModule>();
        var camGo = new GameObject("Cam");
        camGo.transform.SetParent(_root.transform, false);
        _cam = camGo.AddComponent<Camera>();
        _cam.orthographic = true;
        _cam.orthographicSize = 960f; // half the 1920 design height
        _cam.aspect = 1080f / 1920f;
        _cam.transform.position = new Vector3(0f, 0f, -100f);
        _cam.farClipPlane = 2000f;
        _cam.cullingMask = 1 && Uilayer;
        var canvasGo = new GameObject("RigCanvas");
        canvasGo.transform.SetParent(_root.transform, false);
        _canvas = canvasGo.AddComponent<Canvas>();
        _canvas.renderMode = RenderMode.WorldSpace;
        _canvas.worldCamera = _cam;
        ((RectTransform)canvasGo.transform).sizeDelta = new Vector2(1080f, 1920f);
        canvasGo.AddComponent<GraphicRaycaster>();
        _sim = new ShopSim();
    }
}
```

```
var viewGo = new GameObject("ShopView");
viewGo.transform.SetParent(_root.transform, false);
_view = viewGo.AddComponent<ShopView>();
_view.Initialize(_canvas, _sim);
SetLayer(_root, UiLayer);
// Host glue in the launcher shape (BattleLauncher parity):
// view event -> handler -> sim op.
_view.AcceptRequested += () =>
{
    _acceptEvents++;
    if (_sim.AcceptOrder()) _acceptDone++;
};
// Launcher state faces the accept flow rides on - the exact
// OnStateChanged wiring BattleLauncher now ships (WaitingAccept
// shows the bubble; Crafting swaps panels and hides it).
_sim.StateChanged += state =>
{
    switch (state)
    {
        case ShopSim.ShopState.GuestArriving:
            _view.HideOpeningOverlay();
            break;
        case ShopSim.ShopState.WaitingAccept:
            _view.ShowBubble(true);
            break;
        case ShopSim.ShopState.Crafting:
            _view.ShowCraftingPanel(true);
            _view.ShowBubble(false);
            break;
    }
};
_acceptBtn = FindButton("\u63a5\u5335");
// Ledger hygiene (P02 N3 law): NUnit can reuse one fixture
// instance across the tests in this fixture, so every counter
// the roam asserts on is zeroed per test.
_acceptEvents = 0;
_acceptDone = 0;
_materialEvents = 0;
_investDone = 0;
// Launcher parity (BattleLauncher.OnMaterialTapped): a drawer
// seat's tap routes into the sim invest op through the view
// event - the seam the N3 roam proves with a real finger.
_view.MaterialTapped += id =>
{
    _materialEvents++;
    if (_sim.TryInvest(id)) _investDone++;
};
// Discovered seat inventory for the N3 roam: the accept seat,
// the material drawer and the HUD affordances the launcher
// ships. Live or dead, the pointer chain's active/interactable
```

```

        // gate decides what the finger does - never the bot.
        _matButtons = new Button[NewbieConfig.MaterialCount];
        for (int i = 0; i < NewbieConfig.MaterialCount; i++)
            _matButtons[i] = FindButton("Mat_" + i);
        _storyBtn = FindButton("StoryButton");
        _caravanBtn = FindButton("CaravanButton");
        _codexBtn = FindButton("CodexButton");
        _facilityBtn = FindButton("FacilityButton");
        _signBtn = FindButton("SignInButton");
    }
    public override void TearDown()
    {
        if (_root != null) Object.DestroyImmediate(_root);
        _root = null;
        _canvas = null;
        _cam = null;
        _sim = null;
        _view = null;
        _acceptBtn = null;
        _matButtons = null;
        _storyBtn = null;
        _caravanBtn = null;
        _codexBtn = null;
        _facilityBtn = null;
        _signBtn = null;
        base.TearDown();
    }
    // —— the accept seam (VirtualUserCheck persona Tap) through the
    // real pointer chain. Two faces: (1) the live seat at WaitingAccept
    // must SHIP and a synthesized tap at its runtime screen point must
    // land the accept exactly like a finger; (2) once the order is taken
    // the bubble hides and the SAME tap at the SAME screen point must be
    // swallowed by the pointer chain - the exact active/visibility gate
    // the old onClick.Invoke() seam bypassed (it fired the delegate on an
    // inactive GameObject without the engine ever seeing a pointer).
    // ——
    [UnityTest]
    public IEnumerator AcceptFlow_InjectedTouch_WalksRealPointerChain()
    {
        Assert.NotNull(_acceptBtn, "the view must build the accept seat");
        // Walk the shop to WaitingAccept the way the editor bot does
        // (sim ticks; the StateChanged glue shows the bubble on arrival).
        int guard = 0;
        while (_sim.State != ShopSim.ShopState.WaitingAccept && guard++ < 600)
            _sim.Tick(0.1f);
        Assert.AreEqual(ShopSim.ShopState.WaitingAccept, _sim.State,
            "the tutorial shop reaches the accept moment");
        Assert.IsTrue(_acceptBtn.gameObject.activeInHierarchy,
            "the accept seat SHIPS at WaitingAccept - the X1587 soft-lock face");
        Assert.IsTrue(_acceptBtn.interactable, "the live accept seat is tappable");
    }

```

```

    yield return TouchTap(_acceptBtn);
    Assert.AreEqual(1, _acceptEvents, "the injected touch reached the view event");
    Assert.AreEqual(1, _acceptDone, "the handler routed the accept into the sim op");
    Assert.AreEqual(ShopSim.ShopState.Crafting, _sim.State,
        "the tap walked the whole chain: event -&gt; handler -&gt; sim op (order taken)");
    Assert.IsFalse(_acceptBtn.gameObject.activeInHierarchy,
        "the taken order hides the bubble (launcher Crafting face)");
    // The gate face the old seam could never prove: the same finger
    // tap at the same screen point now hits nothing - the pointer
    // chain swallows it before the onClick delegate is ever reached.
    yield return TouchTap(_acceptBtn);
    Assert.AreEqual(1, _acceptEvents, "the hidden seat swallowed the injected tap");
    Assert.AreEqual(1, _acceptDone, "no second accept happened through the hidden seat");
}
// —— N3 (X1595 design freeze section 2, slice c): the random-roam
// bot law. A seeded roam - not a script: blind points across the
// whole face (placeholder bands, dead corners, off-face screen)
// interleaved with pokes at the discovered seats, every one through
// the real pointer chain. Faces: zero softlock (every poke lands
// inside the frame guard and no unexpected log survives); first
// accept reachable via the true chain (the opening beat taps the
// live accept seat and the order lands); the invest book moves
// under the roaming finger (the drawer seat routes a real invest
// during Crafting); the X1587 seat-swallow law holds for the whole
// roam (the taken order's hidden accept seat never fires a second
// accept no matter how often the roaming finger returns to it).
// ——
[UnityTest]
public IEnumerator N3_RandomRoamBot_ZeroSoftlock_FirstAcceptReachable_InvestBookMoves()
{
    const int TotalPokes = 72;    // freeze law: &gt;= 60 pokes on the rig
    const float Dt = 0.05f;      // fixed sim step, editor-check parity
    var rng = new System.Random(20261001);
    // Discovered-seat inventory must ship before any roaming: the
    // accept seat, the drawer and the HUD affordances.
    for (int i = 0; i < _matButtons.Length; i++)
        Assert.NotNull(_matButtons[i], "the drawer seat Mat_" + i + " ships");
    Assert.NotNull(_storyBtn, "the story seat ships");
    Assert.NotNull(_caravanBtn, "the caravan seat ships");
    Assert.NotNull(_codexBtn, "the codex seat ships");
    Assert.NotNull(_facilityBtn, "the facility seat ships");
    Assert.NotNull(_signBtn, "the sign seat ships");
    var seats = new[]
    {
        _acceptBtn, _matButtons[0], _matButtons[1], _matButtons[2], _matButtons[3],
        _storyBtn, _caravanBtn, _codexBtn, _facilityBtn, _signBtn
    };
    // Opening beat precondition: the true state chain carries the
    // rig to the accept moment (Opening -&gt; GuestArriving -&gt;
    // WaitingAccept), exactly the walk the launcher's Update runs.

```

```
int walkGuard = 0;
while (_sim.State != ShopSim.ShopState.WaitingAccept && walkGuard++ < 600)
    _sim.Tick(Dt);
Assert.AreEqual(ShopSim.ShopState.WaitingAccept, _sim.State,
    "the true chain carried the rig to WaitingAccept");
_view.SyncView(_sim);
Assert.IsTrue(_acceptBtn.gameObject.activeInHierarchy,
    "the accept seat SHIPS at WaitingAccept - the X1587 soft-lock face");
int blindPokes = 0, inFaceBlind = 0, offFaceBlind = 0;
int landed = 0; // loop counter - proves the roam never wedged
// The 1080x1920 face rect in screen space, so in-face blind pokes
// can land on placeholder bands (HUD strips, slot rows, counter)
// too, and off-face pokes explore the screen dead zones outside
// the world-space canvas.
var faceMin = RectTransformUtility.WorldToScreenPoint(_cam, new Vector3(-540f, -960f, 0f));
var faceMax = RectTransformUtility.WorldToScreenPoint(_cam, new Vector3(540f, 960f, 0f));
for (int poke = 0; poke < TotalPokes; poke++)
{
    Vector2 point;
    if (poke == 0)
    {
        // Opening beat: the live first-accept seat is inside the
        // roam's poke domain by construction.
        point = SeatPoint(_acceptBtn);
    }
    else if (poke == 3)
    {
        // Drawer beat: a discovered material seat poked while
        // Crafting is live must route a real invest through the
        // pointer chain. Musk (id 3) is the out-of-recipe cold
        // demo - the guided strategy's first invest (L3 parity),
        // so the outcome is deterministic, never RNG-handed.
        point = SeatPoint(_matButtons[3]);
    }
    else if (poke % 3 == 0)
    {
        // Discovered-seat roam - live or dead, the pointer
        // chain's active/interactable gate decides what the
        // finger does, not the bot.
        point = SeatPoint(seats[rng.Next(seats.Length)]);
    }
    else
    {
        blindPokes++;
        bool inFace = faceMax.x > faceMin.x && faceMax.y > faceMin.y && (rng.Next() &
            if (inFace)
            {
                inFaceBlind++;
                point = new Vector2(faceMin.x + (float)rng.NextDouble() * (faceMax.x - faceMin.x),
                    faceMin.y + (float)rng.NextDouble() * (faceMax.y - faceMin.y));
            }
    }
}
```

```

    }
    else
    {
        offFaceBlind++;
        point = new Vector2(rng.Next(Mathf.Max(1, Screen.width)), rng.Next(Mathf.Max(1, Screen.height)))
    }
}
int acceptBefore = _acceptEvents;
int materialBefore = _materialEvents;
yield return TouchTapAt(point);
if (poke == 0)
{
    Assert.AreEqual(1, _acceptEvents - acceptBefore,
        "opening beat: the live seat routed the finger to the view event");
    Assert.AreEqual(1, _acceptDone, "the first accept landed through the real pointer chain");
    Assert.AreEqual(ShopSim.ShopState.Crafting, _sim.State,
        "first accept reachable: the order is taken at the opening beat");
}
if (poke == 3)
{
    Assert.GreaterOrEqual(_materialEvents - materialBefore, 1,
        "drawer beat: the discovered material seat routed the finger to the view event");
    Assert.GreaterOrEqual(_investDone, 1,
        "the invest book moved under the roaming finger (real pointer chain)");
    Assert.GreaterOrEqual(_sim.ColdInvests + _sim.WarmInvests + _sim.CorrectInvests, 1,
        "the routed tap landed a real invest in the sim");
}
// The shop clock runs under the roaming finger (fixed step
// keeps the roam deterministic across machines).
for (int f = 0; f < 8; f++)
{
    _sim.Tick(Dt);
    yield return null;
}
_view.SyncView(_sim);
landed++;
}
Assert.AreEqual(TotalPokes, landed,
    "zero softlock: the roam completed every poke inside the frame guard");
Assert.GreaterOrEqual(blindPokes, 30,
    "blind pokes carried at least a third of the roam (got " + blindPokes + ")");
Assert.Greater(inFaceBlind, 0,
    "both blind domains exercised: in-face placeholder bands (got " + inFaceBlind + ")");
Assert.Greater(offFaceBlind, 0,
    "both blind domains exercised: off-face screen dead zones (got " + offFaceBlind + ")");
Assert.AreEqual(1, _acceptEvents,
    "the X1587 seat-swallow law held the whole roam: the taken order's hidden seat never fired a second acce
Assert.AreNotEqual(ShopSim.ShopState.WaitingAccept, _sim.State,
    "the shop moved past the accept moment under the roaming finger");
LogAssert.NoUnexpectedReceived();

```

```

    }
    // —— rig helpers ——
    Button FindButton(string name)
    {
        foreach (var b in _canvas.GetComponentsInChildren<Button>(true))
            if (b.gameObject.name == name) return b;
        return null;
    }
    /// <summary>Runtime screen point of a seat's center (world-space rig).</summary>
    Vector2 SeatPoint(Button target)
    {
        var rt = (RectTransform)target.transform;
        return RectTransformUtility.WorldToScreenPoint(_cam, rt.position);
    }
    /// <summary>One synthesized finger tap at an arbitrary screen point -
    /// the roam variant of TouchTap (down, one frame, up, two settle frames).</summary>
    IEnumerator TouchTapAt(Vector2 screenPos)
    {
        Assert.Greater(Screen.width, 0, "player has no screen face in this run");
        Assert.Greater(Screen.height, 0, "player has no screen face in this run");
        BeginTouch(1, screenPos);
        yield return null;
        EndTouch(1, screenPos);
        yield return null;
        yield return null;
    }
    static void SetLayer(GameObject root, int layer)
    {
        root.layer = layer;
        foreach (Transform tr in root.GetComponentsInChildren<Transform>(true))
            tr.gameObject.layer = layer;
    }
    /// <summary>One synthesized finger tap at the target's runtime screen
    /// point: touch down, one frame, touch up, then two frames so the
    /// pointer chain (module update -> EventSystem -> onClick) settles
    /// before the caller asserts.</summary>
    IEnumerator TouchTap(Button target)
    {
        var rt = (RectTransform)target.transform;
        var screenPos = RectTransformUtility.WorldToScreenPoint(_cam, rt.position);
        Assert.Greater(Screen.width, 0, "player has no screen face in this run");
        Assert.Greater(Screen.height, 0, "player has no screen face in this run");
        BeginTouch(1, screenPos);
        yield return null;
        EndTouch(1, screenPos);
        yield return null;
        yield return null;
    }
}
}

```